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Primal-Dual Value Function Approximation for Stochastic Dynamic Intermodal Transportation with Eco-Labels

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Abstract. Eco-labels are a way to benchmark transportation shipments with respect to their environmental impact. In contrast to an eco-labeling of consumer products, emissions in transportation depend on several operational factors like the mode of transportation (e.g., train or truck) or a vehicle's current and potential future capacity utilization when new orders are added for consolidation. Thus, satisfying eco-labels and doing this cost efficiently is a challenging task when dynamically routing orders in an intermodal network. In this paper, we model the problem as a multiobjective sequential decision process and propose a reinforcement learning method: value function approximation (VFA). VFAs frequently simulate trajectories of the problem and store observed values (violated eco-labels and costs) for states aggregated to a set of features. The observations are used for improved decision making in the next trajectory. For our problem, we face two additional challenges when applying a VFA, the multiple objectives and the "delayed" realization of eco-label satisfaction due to future consolidation. For the first, we propose different feature sets dependent on the objective function's focus: costs or eco-labels. For the latter, we propose enhancing the suboptimal decision making and observed pessimistic *primal* values within the VFA trajectories with optimistic *dual* decision making when all information of a trajectory is known ex post. This enhancement is a general methodological contribution to the literature of approximate dynamic programming and will likely improve learning for other problems as well. We show the advantages of both components in a comprehensive study for intermodal transport via trains and trucks in Europe.

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Keywords: intermodel transportation • sustainable transportation • eco-label • multiobjective stochastic dynamic decision making • approximate dynamic programming • value function approximation • primal-dual VFA • objective-specific feature selection

1. Introduction

Transportation of freight and passengers is a main contributor to the anthropogenic greenhouse gas emissions (European Environment Agency 2021), and a rising public and political awareness of it forces logistics companies to offer more sustainable transport solutions. A popular method for benchmarking products, services, or facilities according to their environmental impact is to use eco-labels (Lorenzo-Toja et al. 2016). Such labels generally indicate the relative environmental impact according to a preset performance scheme using an intuitive traffic-light coloring system. Recent research has shown that customers are preferably selecting eco-friendly delivery options (Agatz, Fan, and Stam 2021), and consequently, first approaches were made to transfer the concept of eco-labels to transportation (Heinold and Meisel 2020, Kirschstein et al. 2022). Figure 1(a) illustrates such a scheme for the transportation sector using eco-labels A, B, and C. Logistics companies can then let customers specify a desired eco-label for an order or communicate an order's achieved label to gain a competitive advantage. Either way, the emissions promised through eco-labels then ought to be met in the transport operations and shippers need to consider these order-specific guarantees imposed by eco-labels in their decision making.

In contrast to an eco-labeling of consumer products (e.g., for the annual electricity consumption of a refrigerator),

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emissions in transportation depend on several operational factors such as freight consolidations and a vehicle's capacity utilization (Demir, Bektaş, and Laporte 2011, Heinold 2020). For example, Figure 1(b) shows that a freight train's emission rate per ton-km decreases, the more load the train carries; that is, the emission rate and the payload weight are negatively correlated. Next to the high cost pressure in the transportation industry, this emphasizes the need for consolidating freight orders that is particularly relevant in networks with multiple transport modes. Moreover, such intermodal problems often deal with transport demand that reveals over time and requires a decision making under uncertainty regarding the consolidation of freight (Rivera and Mes 2017a). Therefore, decision makers dispatching orders over time need to find a balance between complying with orders' eco-labels as specified by customers and the resulting overall transportation costs.

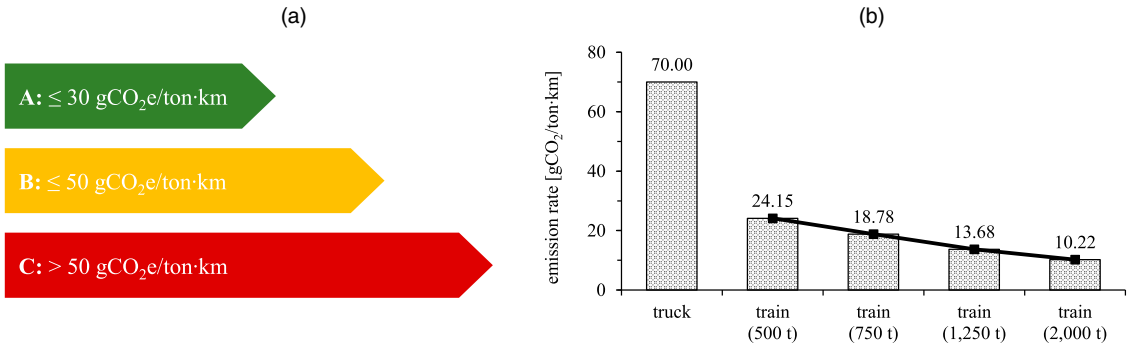
In this paper, we address the challenge of balancing costs and satisfied eco-labels for a stochastic and dynamic decision-making problem in intermodal railroad transportation. We consider a train with limited capacity operating on a fixed schedule where transport orders consisting of full truckloads appear spontaneously over time and request a certain eco-label. Parts of the train's capacity are blocked for orders of long- or midterm contracts but empty capacity can be used for short-term orders made in real time (Railnet Europe 2021). This spot market demand is a general trend in the logistics sector as is apparent by various online platforms for booking freight capacity like Quicargo or Uber Freight, some of them specialized on rail freight like e-Cargo from the Rail Cargo Group or RAILVIS. In our setting, it needs to be decided for each order whether to route it intermodally using the train for the long-haul carriage (and if so, how) or directly using trucks only. The decisions impact costs and the percentage of achieved eco-labels, which are considered in a weighted objective function.

Finding effective solutions for this problem is difficult in two ways. First, we must make the routing decisions for already known orders, and second, we have to anticipate a decision's impact on the future for which orders are not known yet. To tackle this, we propose to use a mixed-integer programming approach in combination with value function approximation (VFA) with basis functions, a solution class from approximate dynamic programming (Powell 2011). The linear basis functions use so-called features whose weights are learned in VFA iteratively in a forward programming fashion using simulated outcomes. The features also work as a parametric representation of the large problem space inherent to the considered problem. For the decision making in VFA, we formulate a mixed-integer linear program that incorporates the basis functions. This method needs to cope with two aspects specific to our setting: (i) learning the feature weights is difficult as the achieved eco-label of an order depends on its own routing and the consolidation decisions for future orders; that is, it is only known for an order once it arrives at its destination as the calculation of emissions requires full knowledge of the train's payload for each relevant service leg; and (ii) we deal with a multiobjective problem considering eco-labels and costs.

To address the learning challenges, we propose a learning strategy that enhances the trajectory obtained from the suboptimal forward decisions in the VFA algorithm (*primal*) with the trajectory obtained from solving each iteration's distinct decision problem to optimality ex post once all information is known (*dual*). We refer to this approach as *primal-dual* VFA. To the best of our knowledge, we are the first to combine both trajectories in the learning process of a transportation problem. To address the multiobjective nature of the problem, we design objective-specific feature sets that take up the factors influencing eco-labels and costs. Overall, our paper makes contributions to the problem as well as to the general methodology on approximate dynamic programming (ADP):

1. We are the first to consider eco-labels in stochastic dynamic intermodal transportation and present a comprehensive multiobjective stochastic dynamic model.

Figure 1. (Color online) Considered Eco-Labeling Scheme and Load-Dependent Emissions Rates



Notes. (a) Eco-labeling scheme. (b) Emission rates from truck and train transportation (EcoTransIT World Initiative 2020).

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2. We present an anticipating solution method considering the unique challenges found in our freight consolidation problem. This method enables effective real-time decision making for our problem.

3. We enhance VFA learning with ex post solutions. This enhancement is a general methodological contribution to the ADP literature and will likely improve VFA for other problems as well.

4. We show the importance and benefits of designing feature sets with respect to the weight in the multiobjective decision problem.

5. We perform comprehensive experiments to analyze the impact of eco-labels on costs and their impact on the embedded decisions in intermodal networks.

The remainder of this paper is organized as follows. Section 2 provides an overview of the related literature and identifies the research gaps addressed in this paper. Section 3 outlines the problem with its distinctive characteristics and formulates it as a sequential decision process. Section 4 motivates and describes our solution approach. Section 5 presents experimental results. Section 6 concludes the paper.

2. Related Literature

We contribute to the literature on multimodal transportation and the literature on VFA. We provide reviews for each of these research directions in Sections 2.1 and 2.3, respectively. In addition, we provide in Section 2.2 a short review of cargo capacity management as there are similarities to our paper regarding problem type and solution approach.

2.1. Multimodal Transportation

We use the term multimodal to refer to the potential usage of more than one transport mode and, with this, intermodal studies are considered as well as they describe a particular type of multimodal transportation in which the same transport unit is used for shipping an order. We also consider studies on synchromodal transportation that usually deal with operational issues regarding the synchronization of orders' time windows and modes' schedules (Rivera and Mes 2017b). In this context, this section reviews two problem classes of multimodal transportation: (i) multimodal transportation problems that consider environmental aspects and (ii) multimodal transportation problems that take place in a stochastic and dynamic environment. There are only few papers considering aspects of both classes. We refer to Crainic and Kim (2007) and Steadie Seifi et al. (2014) for general reviews on multimodal transportation and to Delbart et al. (2021) for a review on uncertainty in intermodal and synchromodal transportation. The very recent survey of Archetti, Peirano, and Speranza (2022) also emphasizes that investigating multiobjective problems with environmental aspects (such as carbon emissions) are a valuable topic for future research. Table 1 lists multimodal papers that are closely related to our study. The table indicates for each paper if it considers stochastic and/or dynamic elements, if it treats emissions as part of the objective function and/or as a restriction in the mathematical model, and, if it relates to multimodal, intermodal, or synchromodal transportation.

Table 1. Overview of Multimodal Studies Related to Our Problem

	Emissions				
Reference	Stochastic	Dynamic	Objective	Restriction	Type
Bauer, Bektaş, and Crainic (2010)	—	—	√	—	Intermodal
Heinold and Meisel (2019)	—	—	√	—	Intermodal
Rudi et al. (2016)	—	—	√	—	Intermodal
Sun and Lang (2015)	—	—	√	—	Multimodal
Baykasoğlu and Subulan (2016)	—	—	√	—	Intermodal
Lam and Gu (2016)	—	—	—	√	Intermodal
Chen, Yang, and Notteboom (2014)	—	—	—	√	Intermodal
Heinold and Meisel (2020)	—	—	√	√	Intermodal
Demir et al. (2016)	√	—	√	—	Intermodal
Zhao, Xue, and Zhang (2018)	√	—	√	—	Intermodal
Jiang et al. (2020)	√	—	—	√	Multimodal
Mes and Iacob (2016)	√	—	√	—	Synchromodal
Resat and Turkay (2019)	√	—	√	—	Synchromodal
Hrušovský et al. (2020)	√	—	√	—	Intermodal
Li, Negenborn, and De Schutter (2015)	√	√	—	—	Intermodal
Rivera and Mes (2017a)	√	√	—	—	Intermodal
Rivera and Mes (2019)	√	√	—	—	Synchromodal
Van Heeswijk et al. (2018)	√	√	√	—	Intermodal
Guo et al. (2021a)	√	√	√	—	Synchromodal
Guo et al. (2021b)	√	√	√	—	Synchromodal
Guo, Atasoy, and Negenborn (2022)	√	√	√	—	Synchromodal
This study	√	√	√	√	Intermodal

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As a first problem class, we consider environmentally oriented studies in multimodal networks. The environmental impact of transportation is usually measured in terms of the resulting emissions and then modeled either as objective or as restriction. When considered as objective, most studies compare the emission minimizing solutions with solutions using other, more traditional, objectives, such as transit times (Bauer, Bektaş, and Crainic 2010, Heinold and Meisel 2019) or costs (Rudi et al. 2016). In the online appendix, we show that our method can also be applied in settings using overall emissions as part of our objective. Emissions can also be considered in traditional objectives, for example, by converting emissions into a monetary value (Sun and Lang 2015) or by setting preset values to each of the potentially conflicting objectives in a goal programming approach (Baykasoğlu and Subulan 2016). Studies that strictly restrict emissions are less common. Lam and Gu (2016) study the tradeoff between costs and time while considering an overall network wide emission limit as a hard constraint. The authors argue that such a limit can result from environmental regulations imposed by a government. Similar approaches consider so-called emission reduction targets that state a reduction of the as-is network emissions. This approach is, for example, used in Chen, Yang, and Notteboom (2014) for a multimodal coastal liner routing problem. A recent study from Heinold and Meisel (2020) introduces emission limits to reflect customers' environmental preferences regarding the transportation of orders in an intermodal rail/road network. Thereby, an individual emission limit is stated for each order, expressing the maximal amount of transport-induced emissions that can be allocated to a particular order. The authors treat these limits both as a hard and as a soft constraint and solve the problem hierarchically with different objectives such as costs, emissions, and number of orders that are routed within their emission limit. Using emission limits is closely connected to using our study's concept of eco-labels. For example, it is possible to convert an order's total emission limit (measured in gram of carbon-dioxide equivalent emissions (gCO_2 e)) to a relative eco-label (measured in gCO_2 e/ton·km) by taking into account the order's payload weight (in ton) and its direct distance (in km) between the origin and the destination location.

Thus far, all the aforementioned studies cover problems in deterministic and static environments. There are only a few studies that deal with uncertainty in the context of environmentally oriented multimodal transportation. Demir et al. (2016) propose a green service network design problem in which the demand as well as the travel times of vehicles are uncertain. Emissions are considered next to time and costs in a weighted multiobjective mixed-integer linear program, which is then applied to several artificial and real-world test instances. Zhao, Xue, and Zhang (2018) consider uncertain demand and supply in an intermodal empty container repositioning problem. Here, the authors explicitly address emissions as part of a weighted cost function and model the problem as a chance-constrained nonlinear integer program. Jiang et al. (2020) consider demand uncertainty in a multimodal network design problem with emission reduction targets. Research on synchromodal transportation usually considers uncertainty for the vehicles' travel times or as unexpected events that then lead to disruptions. Mes and Iacob (2016) introduce a synchromodal planning algorithm allowing decision makers to select from a restricted list of multimodal routes. The authors consider objectives costs, delays, and emissions and apply the approach to the problem of a logistics service provider in the Netherlands. Resat and Turkay (2019) present a discrete-continuous optimization model to solve complex synchromodal routing problems including decisions on the routing and fleet size. Pareto optimal solutions for objectives costs, travel time, and emissions are obtained by applying the ϵ-constraint method. Hrušovský et al. (2020) propose a decision support methodology that connects ex ante offline planning (before the transport has started) with online replanning (in case of disruptions during the transport). With this, the problem also addresses aspects of synchromodal transportation and is then solved for objectives costs, time, and emissions.

The second problem class considers multimodal transportation problems with stochastic *and* dynamic elements, that is, settings in which multiple decision points are considered over time. As far as is known, there is no study in this class that considers environmental aspects by eco-labels. Moreover, there are only a few studies that explicitly address stochastic dynamic problems in multimodal networks. For example, Li, Negenborn, and De Schutter (2015) approach a dynamic intermodal freight transport problem in which demand and traffic conditions are both uncertain. The authors use a problem formulation with a terminology similar to the one that is also used in sequential decision processes. A preceding horizon control algorithm is proposed as solution method and its performance is demonstrated in a simulation study. Rivera and Mes (2017a) study an intermodal freight selection problem in which it is central to assign containers either to a long-haul round trip with a barge or to a direct trip with a truck. The problem spans over multiple periods and, in each period, an uncertain container demand appears at an inland terminal that needs to be transported to deep sea terminals, using one of the aforementioned transport options. The problem is modeled as a Markov decision process (MDP) and solutions are obtained with VFA. Rivera and Mes (2019) also consider an integrated scheduling of drayage and long-haul operations in which the latter is modeled as an MDP and solved via VFA. The aforementioned multimodal stochastic dynamic transportation

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studies do not consider sustainability aspects in their analyses. In contrast, Van Heeswijk et al. (2018) use a multiobjective function considering costs and monetized emissions in an intermodal freight consolidation problem with reloads. The focus is on shipments of small volume (less-than-truckload) that are then consolidated to container loads at the intermodal hubs. Similar to Mes and Iacob (2016), the authors use an arc expansion approach to generate a restricted list of promising transport routes and then identify and evaluate corresponding consolidation opportunities. There is also research on stochastic and dynamic synchromodal transportation. For example, Guo et al. (2021a) consider a hinterland shipment problem in which spot requests appear dynamically over a finite time horizon and need to be matched with multimodal services. The authors minimize costs of the obtained solutions and propose a rolling horizon framework with sample average approximations to deal with the complex decision problem at each stage. This problem setting is extended by stochastic travel times in the works of Guo et al. (2021b) and Guo, Atasoy, and Negenborn (2022). There, the problem is solved under the objective of maximizing profit but other performance measures like delays of deliveries are considered as well. Furthermore, these papers address environmental aspects by considering emission-related costs. To solve the problem, Guo et al. (2021b) use a rolling horizon framework in which a sample average approximation method is used to deal with the stochasticity regarding the spot requests and a chance-constrained programming method to deal with the stochasticity regarding travel times. In contrast, Guo, Atasoy, and Negenborn (2022) develop a reinforcement learning approach and show that this approach improves total profits by 3.37% compared with the approach of Guo et al. (2021b).

Our paper addresses the previously discussed problem classes by investigating a stochastic dynamic intermodal transportation problem with order-specific eco-labels. To comply with the specified eco-labels, the corresponding maximal amount of emissions allowed for an order is (softly) restricted in the underlying model. From this, we consider costs and the fulfillment of eco-labels as two objectives in a multiobjective problem setting, adding to the recently identified need to think beyond the pure cost optimization that is often present in transportation science (Agatz, Hewitt, and Thomas 2021). We also present results on minimizing overall emissions in the online appendix.

2.2. Cargo Capacity Management

In cargo capacity management, reject and accept decisions are considered in the presence of uncertain demand, as is investigated at the example of an airline carrier by Levina et al. (2011) and Levin, Nediak, and Topaloglu (2012). For example, Levin, Nediak, and Topaloglu (2012) consider an airline carrier's cargo capacity that can either be used by requests from allotment contracts or by requests appearing on the spot market. These problems have some similarities with our setting. First, central to our problem is a train that, like an aircraft, does not have sufficient capacity to transport all requests. However, we assume that the carrier must transport orders that do not fit on the train via trucks, and consequently, we focus on costs rather than profits because all cargo will be transported in the end. Moreover, the concept of eco-labels is central to our paper and has not been considered in cargo capacity management. Second, we consider a problem in which orders occur while the train traverses its route, which is similar to spot market demand. The issue of allotment contracts is covered in our paper by reserving a fixed share of the train's capacity for this demand. With this, we focus on the spot market decision and refrain from looking at the interrelated optimization problems inherent when considering a simultaneous optimization for the allotment contracts as well as spot market demand. Third, we model our problem as a sequential decision process as which the dynamic elements in cargo capacity management might be modeled as well (Dumouchelle, Frejinger, and Lodi 2021). We then solve the dynamic problem using VFA and integrate linear approximations directly in a mixed integer linear program (MIP). This integration of linear VFAs is also used in the allotment contract decision in cargo capacity management (Levin, Nediak, and Topaloglu 2012) and many other problems in which decisions depend on solving costly MIP problems (Topaloglu and Powell 2006, Maxwell et al. 2010, Rivera and Mes 2017a). The next section discusses VFA in more detail.

2.3. Improved VFA

VFA is one of the solution classes being available in approximate dynamic programming (Powell 2009, 2011) and has been gained increasing attention in recent years, particularly in the emerging research on stochastic dynamic transportation in unimodal networks (Ulmer et al. 2019, Van Heeswijk, Mes, and Schutten 2019, Ulmer 2020, Ulmer and Savelsbergh 2020, Ulmer et al. 2020). The general idea in VFA is to learn a "good" decision policy from a large number of problem trajectories that are computed with simulation and forward decisions. Thereby, after each (new) trajectory, the policy is updated such that it becomes better and better with each run and, eventually, the policy can be used to generate good solutions within a relatively short online computing time in real-world applications. More details on the principles of VFA are provided in Section 4.

Learning in VFA is especially difficult in the algorithm's first iterations as its initialization is often based on a myopic policy. At the same time, the performance of VFA depends on an effective learning at the beginning of the algorithm as early approximations provide the foundation for future updates; that is, it is important to bring approximations "on the right track." Consequently, the function used in the updating process has

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received special attention as it is a vital determinant for speed and quality of VFA's learning process (George and Powell 2006, Schaul, Zhang, and LeCun 2013, Smith 2017). In this context, Ulmer and Thomas (2020) discuss differences in how to aggregate and store the value of a state. The authors compare parametric VFA, which enables a fast but inaccurate learning, with nonparametric VFA, which leads to accurate approximations but is computationally expensive and therefore slow in the solution process. The authors then propose a meso-parametric VFA that aims at exploiting the benefits of both parametric and nonparametric VFA. Their approach is demonstrated on a capacitated customer order acceptance problem with stochastic requests. George, Powell, and Kualkarni (2008) propose using VFA with multiple lookup tables each at a different level of aggregation. The authors demonstrate the concept of using two levels of aggregation, one rough and one detailed level, and discuss the "optimal weight" of considering the information stored in the two tables for the estimation. This means that an estimate is not obtained from using information from one lookup table (as in traditional VFA) but rather from using information from two (or even more) lookup tables. In doing so, good estimates are obtained by putting a higher weight on the rough lookup table in early iterations and a higher weight on the detailed lookup table in later iterations. However, the level of aggregation for the lookup tables might also result in the classical "exploration versus exploitation" tradeoff. This is especially relevant for detailed lookup tables for which the VFA needs to balance the exploration of values in not yet known areas and the exploitation of values in known areas of the lookup table. Ulmer, Mattfeld, and Köster (2018) present a dynamic lookup table approach that focuses on the learning at the early iterations of the VFA algorithm. In particular, the authors propose a method in which the updating increases in its granularity over the learning process. With this, rough approximations are obtained in the beginning that then become more accurate while the algorithm progresses over further iterations. The method is evaluated on a dynamic vehicle routing problem with stochastic customer requests.

There is also general methodological work on improving VFAs or using primal and dual solutions for dynamic decision making. Jiang and Powell (2015) improve the updating process in VFA by assuming and exploiting monotonicity of the value function. This assumption is reasonable for applications where values decrease over time or remaining resources being available (Ulmer, Mattfeld, and Köster 2018). The authors propose an updating function that explicitly considers the monotone structure by prescribing observed values to unexplored areas of the value function. A faster and better learning of their approach is demonstrated on three problem domains and compared with several other approaches from approximate dynamic programming and reinforcement learning.

Very recently, Jiang, Al-Kanj, and Powell (2020) and Shar and Jiang (2020) proposed the idea of using primal and dual bounds in a Monte Carlo tree search (MCTS) and in Q-learning (QL), respectively, based on the general work from Brown, Smith, and Sun (2010) on information relaxation in stochastic dynamic programs. MCTS is an online lookahead method, where states are evaluated via simulation. In Jiang, Al-Kanj, and Powell (2020), the simulated primal outcome is combined with an ex post solution to evaluate a decision. However, the values are not used for learning. Shar and Jiang (2020) approximates state values as well as upper and lower bounds simultaneously via QL. QL works similar to VFA as it iteratively explores the large-dimensional state space by simulation. However, it evaluates state-action pairs instead of postdecision states, which becomes challenging when the decision space increases. Shar and Jiang (2020) use the lower and upper bounds for penalizing decisions within the learning trajectories and show the superiority of their approach for several toy problems with relatively small decision and state spaces. Our approach shows similarities in using the primal and dual values for guiding decision making and learning. However, we directly integrate the bounds into the update of the VFA. Furthermore, we analyze the performance of our method for a real-world size transportation problem with a very large state space and a combinatorial decision space.

3. Stochastic Dynamic Intermodal Transportation with Eco-Labels

This section describes the stochastic dynamic intermodal transportation problem with eco-labels. Section 3.1 presents an application-oriented outline of the problem, Section 3.2 provides an illustrative example, and Section 3.3 formally describes its sequential decision process.

3.1. Problem Description

We consider a problem where full truck load (FTL) orders dynamically occur over a time horizon and are shipped long-haul via two different modes of transportation. For this, we define the sequence of decision points as $\mathcal{T} = [1, 2, \ldots, T]$. This sequence corresponds to the set of transshipment terminals visited by the train along its route such that one is the first stop and T is the last stop. The set of orders being known at decision point $t \in \mathcal{T}$ is denoted with $\mathcal{O}_t$. Each order $o \in \mathcal{O}_t$ comprises an individual origin, a destination, a payload q_o (measured in FTL), and an *eco-label* λ_o. The eco-label indicates the permissible emissions per ton-km the order is allowed to cause. It expresses the shipper's individual environmental preference. Thereby, emissions

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are measured in terms of equivalents of carbon dioxide (CO_2 e) per ton (t) and per kilometer (km) and, with this, the term emissions subsumes several relevant greenhouse gases, such as carbon dioxide (CO_2), nitrogen oxides (NO_x), sulfur dioxide (SO_2), and nonmethane hydrocarbons (NMHCs) (Piecyk, Cullinane, and Edwards 2012). Figure 1(a) shows the eco-labeling scheme used throughout this paper. The scheme results from the distribution of emission rates obtained from a large-scale simulation study of freight shipments in the European rail/road network (Heinold and Meisel 2018) and comprises three different eco-labels A, B, and C with decreasing strictness. An order with eco-label A has to be transported with an emission rate of at most 30 gCO_2 e/ton· km, an order with eco-label B has to be transported with an emission rate of at most 50 gCO_2 e/ton· km, and an order with eco-label C has no preference regarding its emission rate. To check whether an order o violates its requested eco-label λ_o, we calculate an emission limit $\bar{e}_o$. This limit states the highest amount of emissions allowed for this order according to its requested eco-label. We calculate the limit by using order o's load, the direct origin-destination distance, and the emission rate allowed by the corresponding eco-label.

For transportation, the logistics service provider has access to two modes of transportation: direct transport via truck from a shipment's origin to its destination or consolidated transport via rail. The transportation cost of trucks depend on several factors, such as distance, empty returns, or if they are owned by the company or booked via an online freight exchange platform (Miller, Nie, and Liu 2020). Well-known platform examples include Teleroute in France, Echo and Coyote in the United States, 56QQ in China (also known as Truck Alliance), and Uber Freight. Emissions of a truck transportation depend directly on the origin and destination of the order and, as we consider large orders, we assume a constant emission rate per driven vehicle-kilometer corresponding to a regular fully loaded 40-ton truck, depicted in the first bar in Figure 1(b). Regarding costs, we assume that trucks operate with a constant cost rate per kilometer that can differ dependent on the aforementioned factors. For rail transportation, a single, high-capacity train visits a set of predefined stops over the planning horizon. For example, the train starts in Scandinavia, traverses Germany and Austria, and eventually ends its trip in the south of Italy. At each stop along the route, the train can load/unload nearby orders requiring only a short time driving. To use the train, an order must be transported via truck to a nearby stop before the train arrives (precarriage) where it is then loaded onto the train (main carriage) and, eventually, transported via truck from a subsequent stop of the train to the final destination (postcarriage). An order's costs and emissions are an outcome of these carriages. Like for trucks, we consider a constant cost rate per cargo unit for the train. This is because the scheduled train operates independently of the decisions made in our problem, from which we can assume that cargo units can be booked on the train at a fixed price. The emissions for orders transported via train depend on the pre- and postcarriage distances to those train stops that are selected for loading and unloading an order and on the overall load of the train between these two stops. For the train, we assume a degressive emission rate per ton-km that reduces with the train's total payload. Figure 1(b) shows the emission rates assumed for the freight train, which are derived from the emission estimation model of the EcoTransIT World Initiative (2020). It illustrates that the emission rate per ton-km decreases with a higher payload utilization.

For each stop of the train, the service provider decides about the transport of the nearby orders; direct transport via truck, shipment via train from this stop, or postponement of the shipment decision in case the order can also be loaded at a future stop of the train. We refer to the online appendix for an illustration of an order's corresponding possible routings. Once an order is decided to be loaded onto the train, trucks are booked for the pre- and postcarriage and the customer is informed, implying that the routing of this order cannot be changed in future periods. Thus, a routing decision for an order is final only once it is loaded onto the train's current stop or the train passes the order's last possible loading stop. The routing of the order is then fixed for the remainder of the time horizon as a truck must be booked for the postcarriage.

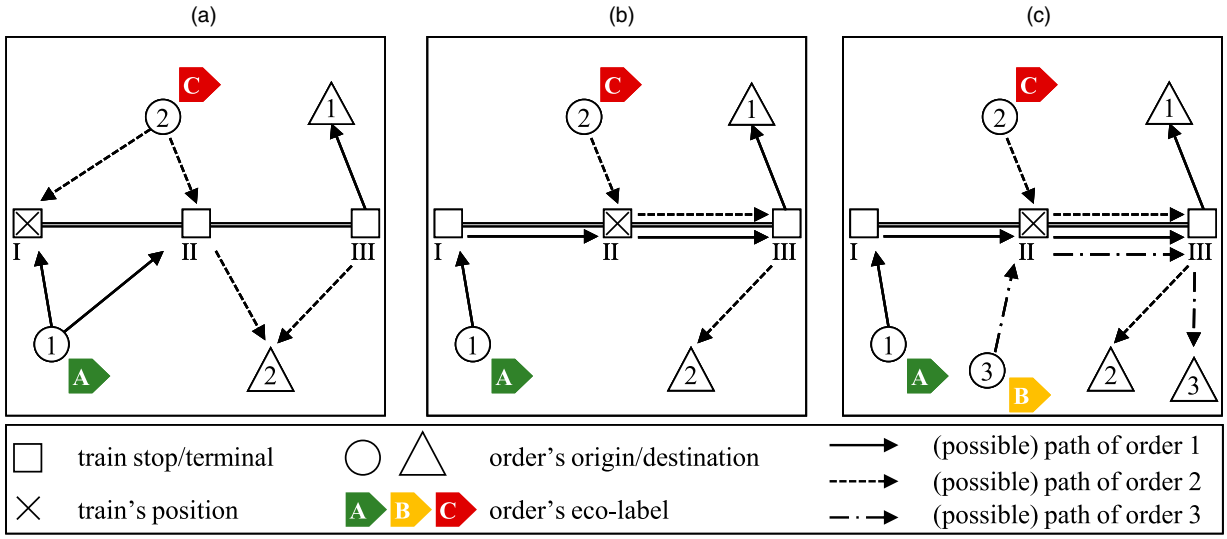
While the train traverses along its route, new orders are placed by the customers that leads to a stochastic and dynamic planning environment. When making a decision, the service provider has two objectives: minimizing the overall transportation costs and minimizing the number of orders that violate their requested eco-labels. In contrast to eco-labels in other domains, for example, for electronics or buildings, the final emissions per order are not yet known when a transport decision is to be made. This is because they depend on the consolidation effects of the train with already loaded orders and with future loads at later stops. Thus, the emissions that are to be allocated to an order and the actually achieved eco-label only become known once the order arrives at its destination.

3.2. Illustrative Example

We provide an illustrative example of the resulting stochastic dynamic intermodal transportation problem with eco-labels. We use here the terms of the sequential decision process that is later formalized in the next section. Figure 2(a) shows a state, Figure 2(b) shows for a potential decision the resulting postdecision state, and Figure 2(c) shows the next state after a transition. Squares represent the three terminals that the train visits in this example where a cross inside a square indicates

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Figure 2. (Color online) Stochastic Dynamic Intermodal Transportation Problem with Eco-Labels



Notes. (a) State. (b) Postdecision state. (c) Next state.

the current stop of the train. Circles represent the origin locations of the three orders that occur in the course of time and triangles represent their destinations. The eco-label requested for an order is shown besides its origin location. Potential pre- and postcarriages via truck are shown as arcs. Direct truck transportation is not depicted in the figure for reasons of simplicity. We assume that the train is empty at the beginning and has a sufficiently large capacity to jointly transport the three orders.

In Figure 2(a), the train just arrived at the first terminal. Here, orders 1 and 2 have been placed by the customers and can be loaded onto the train at the train's current stop at terminal I or at its next stop at terminal II. Order 1 can be unloaded at terminal III only and order 2 can be unloaded at terminals II or III. Thus, there are three possible paths for order 1 (loading at I and unloading at III; loading at II and unloading at III; direct shipping) and four possible paths for order 2 (loading at I and unloading at II; loading at I and unloading at III; loading at II and unloading at III; direct shipping). This illustrates that already small intermodal networks can contain a substantial number of transport options. In general, if an order has n terminals for being loaded and m terminals for being unloaded, we face up to $(n \times m) + 1$ paths for the case that the two sets of terminals are disjunct. Figure 2(b) describes a feasible decision in which orders 1 and 2 are loaded onto the train at terminals I and II, respectively, and unloaded at terminal III. Such a decision results from considering objectives costs and eco-labels. For example, order 1 might be assigned to the first terminal as its requested eco-label A requires as much rail transportation as possible. In contrast, the eco-label of order 2 is only C and the more nearby terminal II might be selected as this results in lower trucking costs (compared with a routing via terminal I). After making these decisions, the train moves to the next station and, with this, Figure 2(b) represents the corresponding postdecision state. Figure 2(c) depicts the next state in which new information has become available. In particular, the new order 3 appears with its corresponding possible intermodal path: loading at terminal II and unloading at terminal III. Furthermore, the set of paths of all other orders is updated in the transition: selected paths including stops that have been visited before are fixed (in this example, the path for order 1 that required a loading at terminal I) and not selected paths including stops that have been visited before are excluded as they can no longer be selected in the new state (in this example, paths for order 2 that required a loading a terminal I).

3.3. Sequential Decision Process

We use the framework of sequential decision processes to formally describe the problem (Powell 2021). We start by introducing some general path notation and then subsequently define states, decisions, rewards, postdecision states, stochastic information, transitions, and the optimal policy. In the online appendix, we provide a provides a complete overview of the notation.

3.3.1. General Path Notation. For each order $o \in \mathcal{O}_t$, we deduce a set of paths $\mathcal{P}_t^o$. This set contains the feasible intermodal paths of an order o (as was illustrated in the example) and a path for the direct transportation via truck between the order's origin and destination. We denote the loading and unloading terminal of an intermodal path $p \in \mathcal{P}_t^o$ by t_p^1 and t_p^2, respectively, and the set of all terminals that are involved in this path by $\mathcal{T}_p$. For

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paths corresponding to a direct (truck only) routing, this set is empty ($\mathcal{T}_p = \emptyset$). The set of all paths being relevant at a decision point t is denoted by $\mathcal{P}_t = \bigcup_{o \in \mathcal{O}_t} \mathcal{P}_t^o$.

3.3.2. State. A state S_t at decision point $t \in \mathcal{T}$ is described by tuple $S_t = (\mathcal{O}_t, \mathcal{P}_t, \boldsymbol{r_t}, \boldsymbol{f_t})$ and contains all information required to make a decision. In particular, $\mathcal{O}_t$ is the set of orders placed until t, $\mathcal{P}_t$ is the set of paths as defined before, $\boldsymbol{r_t} = [r_t(p)]_{p \in \mathcal{P}_t}$ is a $|\mathcal{P}_t|$-dimensional binary vector describing for each path p whether it is part of the current solution ($r_t(p) = 1$), or not ($r_t(p) = 0$), and $\boldsymbol{f_t} = [f_t(o)]_{o \in \mathcal{O}_t}$ is an $|\mathcal{O}_t|$-dimensional binary vector describing for each order o whether it currently violates its eco-label ($f_t(o) = 1$), or not ($f_t(o) = 0$). Vector $\boldsymbol{f_t}$ can be derived from vector $\boldsymbol{r_t}$ (as will be shown in the next paragraph) and is only included in the definition of the state for reasons of readability.

3.3.3. Decision. The set of feasible decisions in a state S_t at decision point $t \in \mathcal{T}$ is described with $\mathcal{X}(S_t)$. A single decision $\boldsymbol{x_t} = [x_t(p)]_{p \in \mathcal{P}_t} \in \mathcal{X}(S_t)$ is defined as a $|\mathcal{P}_t|$-dimensional binary vector describing for each path p whether it is used under the decision ($x_t(p) = 1$), or not ($x_t(p) = 0$). There are two restrictions for a decision to be feasible. First, exactly one path must be selected per order ($\sum_{p \in \mathcal{P}_t^o} x_t(p) = 1, \quad \forall o \in \mathcal{O}_t$) and, second, the capacity of the train $L^{\max}$ cannot be exceeded ($l_t(t') = \sum_{o \in \mathcal{O}_t} (q_o \cdot \sum_{p \in \mathcal{P}_t^o : t' \in \mathcal{T}_p} x_t(p)) \leq L^{\max}, \quad \forall t' \in \mathcal{T}$). Thereby, the load of the train $l_t^x(t')$ at terminal $t' \in \mathcal{T}$ under decision $\boldsymbol{x_t}$ is calculated by summing up the load q_o of all orders o whose selected path involves terminal t'.

3.3.4. Reward. We consider a multiobjective problem and use parameter α to weight objectives costs and eco-labels. A pure cost minimization is achieved by setting $\alpha = 1$ and a pure minimization of the number of orders violating their eco-label results from $\alpha = 0$. We use the term "reward" to describe the contribution of a decision toward each of these objectives. As we have a minimization problem, our goal is to minimize this reward. We then consider marginal rewards instead of absolute rewards (Ulmer et al. 2020) that is motivated by the fact that the actually achieved eco-label of an order can change at later decision points due to the load-dependent emission rate of the train. For example, an order's eco-label might be violated at the time it is loaded onto the train at stop t_p^1 if the train is quite empty at that point, but the eco-label might be fulfilled at the unloading stop t_p^2, if later decisions increase the train's payload utilization and, with this, reduces the emission rate per cargo unit.

To evaluate whether an order $o \in \mathcal{O}_t$ violates its eco-label λ_o, we calculate the emissions of a path $p \in \mathcal{P}_t$ under a decision $\boldsymbol{x_t}$ as $e_t^x(p) = e_p^{truck} + q_o \cdot \sum_{t' \in \mathcal{T}_p \setminus \{t_p^2\}} e^{train}(l_t^x(t')), \quad \forall o \in \mathcal{O}_t, p \in \mathcal{P}_t^o$. Here, the path's emissions resulting from truck transportation are denoted with e_p^{truck}, considering emissions from pre- and postcarriage for intermodal paths and emissions from direct road-only transportation for unimodal paths. The emissions from train transportation per unit of load q_o are denoted by $e^{train}(l_t^x(t'))$ and depend on the train's total payload $l_t^x(t')$ at a terminal $t' \in \mathcal{T}$.

The evaluation of the eco-labels under the current decision $\boldsymbol{x_t}$ is then done for each order using Equation (1). Here, we define $\boldsymbol{f_t^x} = [f_t^x(o)]_{o \in \mathcal{O}_t}$ as a $|\mathcal{O}_t|$-dimensional binary vector describing for each order o whether it violates its eco-label λ_o ($f_t^x(o) = 1$), or not ($f_t^x(o) = 0$), under the current decision $\boldsymbol{x_t}$.

$$f_t^x(o) = \begin{cases} 1, & \text{if } \sum_{p \in \mathcal{P}_t^o} x_t(p) \cdot e_t^x(p) > \bar{e}_o \\ 0, & \text{else,} \end{cases} \qquad \forall o \in \mathcal{O}_t \tag{1}$$

With this, the marginal reward $R(\cdot)$ under a decision $\boldsymbol{x_t}$ is computed by Equation (2). This formula takes into account the marginal costs $R_t^C(\cdot)$ and eco-label violations $R_t^E(\cdot)$ that are calculated as the difference in costs/eco-label violations between the newly planned paths and the previous ones. For this, the marginal costs use c_p to denote the costs of using a path $p \in \mathcal{P}_t$ and are calculated a follows: $R_t^C(S_t, \boldsymbol{r_t}, \boldsymbol{x_t}) = \sum_{p \in \mathcal{P}_t} c_p \cdot (x_t(p) - r_t(p))$. Similarly, the marginal eco-label violations are calculated as $R_t^E(S_t, \boldsymbol{f_t}, \boldsymbol{f_t^x}) = \sum_{o \in \mathcal{O}_t} (f_t^x(o) - f_t(o))$.

$$R(S_t, \boldsymbol{r_t}, \boldsymbol{x_t}, \boldsymbol{f_t}, \boldsymbol{f_t^x}) = \alpha \cdot R_t^C(S_t, \boldsymbol{r_t}, \boldsymbol{x_t}) + (1 - \alpha) \cdot R_t^E(S_t, \boldsymbol{f_t}, \boldsymbol{f_t^x}) \tag{2}$$

3.3.5. Postdecision State. We use the two-step transition in which we first transit to a postdecision state S_t^x considering the decision $\boldsymbol{x_t}$ and afterward to the new state S_{t+1} considering the new information. Thus, the transition to the postdecision state is fully deterministic and formally defined as the state that is reached by making decision $\boldsymbol{x_t}$ in state S_t: $S_t^x = (\mathcal{O}_t, \mathcal{P}_t^x, \boldsymbol{x_t}, \boldsymbol{f_t^x})$. Here, compared with state S_t, $\boldsymbol{r_t}$ and $\boldsymbol{f_t}$ are replaced by the decision $\boldsymbol{x_t}$ and the corresponding eco-label violation vector $\boldsymbol{f_t^x}$, respectively, and the set of available paths $\mathcal{P}_t^x = \bigcup_{o \in \mathcal{O}_t} \mathcal{P}_t^{o,x}$ is updated by excluding all paths that are not feasible in the next stage any more. This exclusion is described for each order's paths $\mathcal{P}_t^{o,x}$ in Equation (3). In particular, according to the first case, we only include the selected path in the postdecision state if it involves loading at the train's current location. From this, once we decide to load an order onto the train at the current stop $t_p^1 = t$, we also fix the unloading terminal t_p^2 according to the selected path p. In the second case, we also only consider the selected path if there is just one path available for this order; that is, the decision is made automatically because of missing alternatives. Otherwise, if none of the previous cases apply, according to the third case in Equation (3), we update the paths such that we only keep those intermodal

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paths p' for which $t^1_{p'} \geq t+1$, simply because paths with loading at earlier train stops are not accessible any more when the train approaches its next stop $t+1$.

$$\mathcal{P}^{o,x}_t = \begin{cases} \{p\}, \text{ if } (x_t(p)=1 \text{ and } t^1_p = t) \\ \{p\}, \text{ if } |\mathcal{P}^o_t| = 1, \qquad \forall o \in \mathcal{O}_t, p \in \mathcal{P}^o_t \\ \{p' \mid p' \in \mathcal{P}^o_t : t^1_{p'} \geq t+1\}, \text{ else} \end{cases} \tag{3}$$

3.3.6. Stochastic Information. The stochastic information Ω in our problem relates to the arrival of new orders and their characteristics. We use $\hat{\boldsymbol{\omega}}_{\boldsymbol{t+1}} \in \Omega$ to describe a potential realization of this information at decision point $(t+1)$, consisting of new orders $\tilde{\mathcal{O}}_{t+1}$ and corresponding new paths $\tilde{\mathcal{P}}_{t+1}$, that is, $\hat{\boldsymbol{\omega}}_{\boldsymbol{t+1}} = (\tilde{\mathcal{O}}_{t+1}, \tilde{\mathcal{P}}_{t+1})$. The distribution functions used in this process are later specified in the experimental section.

3.3.7. Transition. The transition generally describes how the system evolves over time from one state S_t in point t to a next state S_{t+1} in point $t+1$. Using the stochastic information $\hat{\boldsymbol{\omega}}_{\boldsymbol{t+1}}$, we can transit from S^x_t to S_{t+1} using system model function $S^M(S^x_t, \hat{\boldsymbol{\omega}}_{\boldsymbol{t+1}}) = (\mathcal{O}_{t+1}, \mathcal{P}_{t+1}, \boldsymbol{r_{t+1}}, \boldsymbol{f_{t+1}})$. Here, the set of orders is updated as $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \tilde{\mathcal{O}}_{t+1}$, the set of paths is updated as $\mathcal{P}_{t+1} = \mathcal{P}^x_t \cup \tilde{\mathcal{P}}_{t+1}$, the routing vector is updated as $\boldsymbol{r_{t+1}} = [x_t(p)]_{p \in \mathcal{P}^x_t} \cup [0]_{p \in \tilde{\mathcal{P}}_{t+1}}$, and the eco-label violation vector is updated as $\boldsymbol{f_{t+1}} = [f^x_t(o)]_{o \in \mathcal{O}_t} \cup [0]_{o \in \tilde{\mathcal{O}}_{t+1}}$. The default value for the routing and eco-label violation vector is 0.

3.3.8. Optimal Policy. We are searching for a policy $\pi \in \Pi$ describing a decision $\boldsymbol{x_t}$ for each state S_t. Thereby, the set of all policies is denoted with Π and an optimal policy π^* minimizes the objective function from Equation (2) over all of the considered periods. Formally, such a decision can be calculated in our discrete problem by using the well-known Bellman equation (Bellman 1957). The equation considers for a decision $\boldsymbol{x_t}$ in the current state S_t the immediate reward $R(\cdot)$ and the expected future reward of the postdecision state when following an optimal policy. We use $V_t(S^x_t)$ to denote the future reward of a postdecision state S^x_t. The optimal decision $\boldsymbol{x^*_t}$ at decision point t can then be found by solving the following Bellman equation:

$$\boldsymbol{x^*_t} = \underset{\boldsymbol{x_t} \in \mathcal{X}(S_t)}{\arg\min} \left(R(S_t, \boldsymbol{r_t}, \boldsymbol{x_t}, \boldsymbol{f_t}, \boldsymbol{f^x_t}) + \mathbb{E}(V_t(S^x_t)) \right). \tag{4}$$

4. Solution Approach

This section describes our proposed solution approach. Section 4.1 motivates the use of VFA with basis functions by highlighting the challenges of the multimodal problem with eco-labels. Section 4.2 describes the MIP that is solved for a stage's decision problem, and Section 4.3 elaborates on our enhancement regarding the learning of weights of the basis functions. Section 4.4 describes the procedure for selecting the basis functions for the VFA.

4.1. Motivation

As many other stochastic dynamic transportation problems, we face the well-known curses of dimensionality when trying to solve the Bellman equation. For our problem, the decision space is vast due to the different potential paths for the orders in the system. Furthermore, evaluating a decision is challenging as not only future requests and their reward are uncertain but the fulfillment of eco-labels of existing orders may also change based on future requests and decisions.

We address these issues as follows. For the complex decision problem at each stage, we formulate an MIP that can be solved within reasonable computing time by state-of-the-art MIP solvers. This MIP is used to make decisions regarding the routing of the orders in every state. To evaluate decisions, we integrate the Bellman equation in the MIP and approximate the unknown second term of Equation (4) via VFA with basis functions.

To this end, the goal of VFA with basis functions is to replace a postdecision state's S^x_t true value $V_t(S^x_t)$, which is intractable to be calculated for large-dimensional problems, by a good approximation $\bar{V}_t(S_t)$. This approximation is calculated by using a set of features denoted with $\mathcal{A}$. Each feature $a \in \mathcal{A}$ is associated with a basis function $\phi_a(S^x_t)$, describing how the value of feature a is calculated for a postdecision state S^x_t, and a weight θ_a, describing its relative importance for the approximation. With this, the approximation of a postdecision state's value is calculated as follows:

$$V_t(S^x_t) \approx \bar{V}_t(S^x_t) = \sum_{a \in \mathcal{A}} \theta_a \cdot \phi_a(S^x_t). \tag{5}$$

In VFA with basis functions, the values of the weights are learned through a combination of simulation and forward programming over a large number of iterations. We use the superscript n to map values to iterations. A generic VFA algorithm is described in the online appendix.

To enable embedding the VFA into the MIP and using standard solvers, our VFA relies on linear basis functions. Therefore, the states are represented by a set of meaningful features. The purpose of features is to aggregate the high-dimensional (postdecision) state space while allowing differentiation of states with different values. For our multiobjective problem, the value of states depends on both costs and eco-label violations and their weighting parameter α. Some features may mainly reflect future costs as others may solely reflect different values of eco-level violations. Because using a large number of features often obstructs learning of the VFA, we propose using different feature sets for different α-parameter values.

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Another challenge when applying VFA to our problem is that the consequences of current decisions realize substantially later in the process. For example, loading goods on the train in an early state reduces valuable train capacity but also the relative emissions of requests loaded onto the train in the future. We therefore observe a tradeoff between flexibility and consolidation. Furthermore, the number of eco-label violations may be reduced by future consolidation effects. As a VFA uses only approximated values in the (therefore suboptimal) decision making, the observed values in a trajectory are a *primal bound* of the potential values that could have been achieved in the same trajectory. Classical VFAs usually start with an empty value function and therefore myopic decision making (Ulmer and Thomas 2020) with a potentially very weak primal bound, especially for our problem due to the aforementioned reasons. That means that the observed and stored values are likely substantially different compared with the real values. This often leads to poor decision making especially in early trajectories and may even lead to no learning at all. We therefore propose a combination of the too pessimistic primal bound values with *dual bound* values. To this end, we solve the unique decision problem of the primal trajectory recursively ex-post when all information is known. The results are dual bound values, potentially too optimistic compared with the expected values of the observed states. To balance the pessimism of the primal bounds with the optimism of the dual bounds, we update the VFA values with a convex combination of both. We denote this strategy as *primal-dual VFA*. In the following, we will first discuss the general concept of VFAs with basis functions. Then, we will define our primal-dual VFA in detail. Finally, we will present the features we will use for different weights of the α-parameter.

4.2. MIP for a Stage's Decision Problem

The MIP that is solved at each stage is principally a representation of the Bellman equation (4) for a single decision point t using the basis functions to approximate the value of a postdecision state. The MIP is formulated as follows, where the iteration superscript n is dropped for reasons of readability. The constraints of the MIP resemble a decision as described in the sequential decision process.

$$\min\ \alpha \cdot R_t^C(\cdot) + (1-\alpha)\cdot R_t^E(\cdot) + \sum_{a\in\mathcal{A}} \theta_a \cdot \phi_a(\cdot) \tag{6}$$

subject to

$$\sum_{p\in\mathcal{P}_t^o} x_t(p) = 1, \quad \forall o \in \mathcal{O}_t, \tag{7}$$

$$l_t(t') = \sum_{o\in\mathcal{O}_t}\left(q_o \cdot \sum_{p\in\mathcal{P}_t^o : t'\in\mathcal{T}_p} x_t(p)\right) \le L^{\max}, \quad \forall t' \in \mathcal{T}, \tag{8}$$

$$e_t^x(p) = e_p^{truck} + q_o \cdot \sum_{t'\in\mathcal{T}_p\setminus\{t_p^2\}} e^{train}(l_t^x(t')), \quad \forall o\in\mathcal{O}_t, p\in\mathcal{P}_t^o, \tag{9}$$

$$f_t^x(o) = \begin{cases} 1, & \text{if } \sum_{p\in\mathcal{P}_t^o} x_t(p)\cdot e_t^x(p) > \bar{e}_o \\ 0, & \text{else,} \end{cases} \quad \forall o\in\mathcal{O}_t, \tag{10}$$

$$e_t^x(p), l_t^x(t') \ge 0, \quad \forall p\in\mathcal{P}_t, t'\in\mathcal{T}, \tag{11}$$

$$x_t(p), f_t^x(o) \in \{0,1\}, \quad \forall p\in\mathcal{P}_t, o\in\mathcal{O}_t. \tag{12}$$

The first two terms of Objective Function (6) represent Reward Function (2), where objectives costs (R_t^C) and eco-labels (R_t^E) are weighted by a parameter α. The third part of the objective describes the basis functions and reflects the reached postdecision state. They are formulated as a linear combination of the features $a\in\mathcal{A}$ outcome values $\phi_a(\cdot)$, weighted by θ_a. The features and their formulation are later introduced in Section 4.4. Feature weights θ_a are determined by our VFA algorithm and given as input to the MIP. Constraints (7) state that each order must use exactly one of its paths. Constraints (8) define the load of the train at each terminal, which cannot exceed the train's capacity. Constraints (9) calculate the emissions per path p under a decision and are denoted by $e_t^x(p)$. Here, $e^{train}(l_t^x(t'))$ describes train's emissions as a nonlinear function of the train's load. This is easily linearized through further binary decision variables that test for the relevant payload ranges (Figure 1(b)), which is, however, not shown here in the model for reasons of comprehensibility. Constraints (10) set the eco-label violation decision variable $f_t(o) = 1$ if an order o violates its eco-label and $f_t(o) = 0$ otherwise. These nonlinear constraints can be implemented by using big-M formulations or indicator constraints, which are usually provided by state-of-the-art solvers. Constraints (11) and (12) state the domains of the decision variables.

4.3. Learning in Primal-Dual VFA

This section describes how the primal-dual VFA is embedded in the general VFA algorithm described in the previous section. In particular, in primal-dual VFA, the feature weights are updated after each iteration using two trajectories. In the primal trajectory, values of postdecision states are evaluated based on the (suboptimal) forward decisions as in classical VFA. In the dual trajectory, values of postdecision states are evaluated recursively solving an ex post optimization problem using the same stochastic information as in the primal trajectory. That is, a decision is made in each state assuming knowledge about future orders, resulting in a trajectory different from the primal trajectory. With this, our primal-dual VFA relies partly on information that is not available at the point in which the decision is made. This information is used ex post *in the learning*

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process to update the basis functions for decisions in the next stage. As discussed in the literature review, this is one of our method's main novelties.

For the ex post optimization, we solve a deterministic problem at the last decision point, considering *all* orders and paths of iteration n and break down the resulting optimal routings recursively to evaluate what the perfect decision in each decision point $t \in \mathcal{T}$ would have been. In doing so, the dual trajectory constitutes a lower bound for the primal trajectory's minimization problem and provides a valuable piece of information in the learning process. It allows us to pretend a fully deterministic setting for which it is much easier to calculate an optimal solution compared with the multidimensional stochastic setting. In our problem setting, the dual trajectory can be calculated within seconds using a standard solver.

Figure 3 conceptually depicts the primal-dual VFA. Here, states are shown as rectangles, postdecision states as circles, solid black arcs represent the decision making between states and postdecision states, and dashed black arcs the arrival of stochastic information after the postdecision state. The upper part of the figure shows the primal trajectory: it starts at initial state S_1 and ends at final state S_T^{primal} by making decisions $\boldsymbol{x}_t^{\text{primal}}$ at points $t < T$ and by simulating information $\hat{\boldsymbol{\omega}}_{t+1}$ after each decision. Here, decisions are made using the MIP with the integrated current value function, which is depicted through the dotted arcs at the top of the figure. The lower part of the figure shows the dual trajectory: it also starts at initial state S_1 but ends at a final state S_T^{dual}. In the dual trajectory, a decision $\boldsymbol{x}_t^{\text{dual}}$ is made at each decision point and the same information $\hat{\boldsymbol{\omega}}_{t+1}$ from the primal trajectory is used to transit to the next stage. Like in the primal trajectory, decisions are also made using the MIP, but now, the values from the recursive ex post optimization are used for the value function. This is illustrated by the dotted arcs at the bottom of the figure.

Once the primal and dual trajectories are calculated, we update the weights of the features using the states and their respective values from both trajectories. The update of the weights θ_a is formally done by using a primal-dual learning parameter $0 \le \beta \le 1$. This parameter scales the relative importance of each trajectory: $\beta = 0$ means that only the primal trajectory is used to fit the weights, $\beta = 1$ means that only the dual trajectory is used to fit the weights, and $\beta \in (0,1)$ describes combinations thereof. We use a linear regression model for the computation of the weights and integrate the relation expressed through β by weighting the trajectories accordingly. We use a preset value of $\beta = 0.5$ in our experiments but also show results for varying β values in Section 5.

Algorithm 1 (Primal-Dual VFA with Basis Functions)

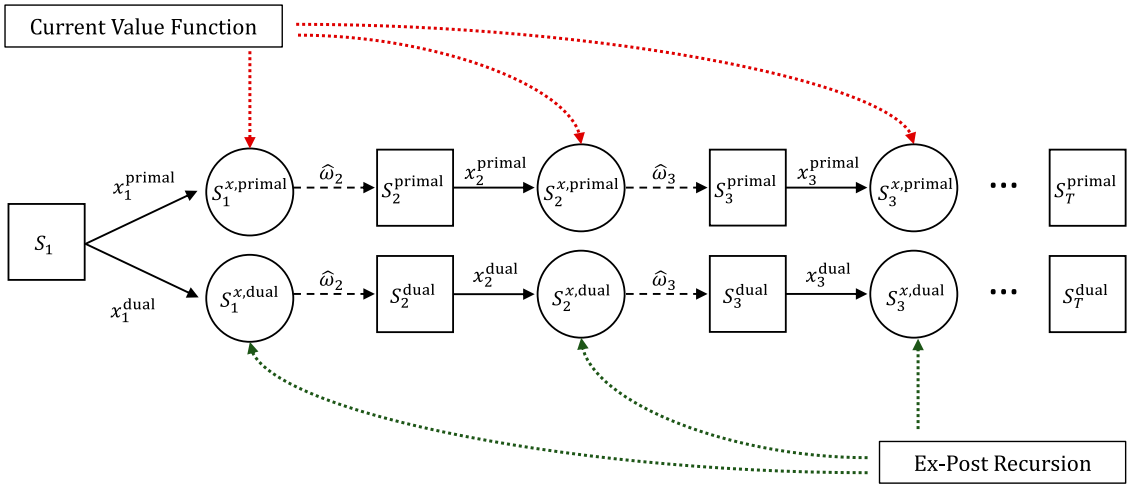
Step 1: Choose initial values for weights θ_a^0 of all features $a \in \mathcal{A}$.
for $n = 1, 2, \ldots, N$ **do**
  Step 2: Draw information $\hat{\boldsymbol{\omega}}_t^n \in \Omega$ for all periods and for both trajectories.

  Primal trajectory
  Step 3a: Use information $\hat{\boldsymbol{\omega}}_1^n \in \Omega$ for initial state $S_1^{n,\text{primal}}$.
  for $t = 1,2,\ldots,T-1$ **do**
    Step 3b: Find primal decision $\boldsymbol{x}_t^{n,\text{primal}}$ by solving MIP Model (6) to (12) using weights θ_a^{n-1}.
    Step 3c: Store decision $\boldsymbol{x}_t^{n,\text{primal}}$, postdecision state $S_t^{n,x,\text{primal}}$, and accumulated reward.
    Step 3d: Use information $\hat{\boldsymbol{\omega}}_{t+1}^n \in \Omega$ to transform to next state: $S_{t+1}^{n,\text{primal}} = S^M(\cdot)$.
  end

  Dual trajectory
  Step 4a: Use information $\hat{\boldsymbol{\omega}}_1^n \in \Omega$ for initial state $S_1^{n,\text{dual}}$.
  for $t = 1,2,\ldots,T-1$ **do**
    Step 4b: Find dual decision $\boldsymbol{x}_t^{n,\text{dual}}$ by solving MIP Model (6) to (12) without the basis functions and assuming that information $\bigcup_{t=1}^{T} \hat{\boldsymbol{\omega}}_t^n$ is known.
    Step 4c: Store decision $\boldsymbol{x}_t^{n,\text{dual}}$, postdecision state $S_t^{n,x,\text{dual}}$, and accumulated reward.
    Step 4d: Use information $\hat{\boldsymbol{\omega}}_{t+1}^n \in \Omega$ to transform to next state: $S_{t+1}^{n,\text{dual}} = S^M(\cdot)$.
  end
  Step 5: Update weights θ_a^n for all features $a \in \mathcal{A}$ using the stored values from Steps 3c and 4c.
end

The general steps of the primal-dual VFA are shown in Algorithm 1. Just like the general VFA that is shown in the online appendix, the primal-dual VFA starts with an initial selection of the feature weights in Step 1. Then, the algorithm processes over N iterations and, for each iteration n, it computes the primal trajectory and the dual trajectory. We again use the superscript n to map values to iterations. The information $\hat{\boldsymbol{\omega}}_t^n$ is drawn in Step 2 for all periods and is then used in both trajectories. In particular, the primal trajectory is computed in Steps 3a–3d, and the same general type of steps are also used to compute the dual trajectory (Steps 4a–4d). In Step 4a, the initial state $S_1^{n,\text{dual}}$ is initialized by using information $\hat{\boldsymbol{\omega}}_1^n$ as was used in the primal trajectory. Then, for each decision point t, a dual decision is found in Step 4b, and it is stored together with the value of the postdecision state and the accumulated reward in Step 4c. For the decision making, there are two important differences compared with the primal trajectory: first, we solve the MIP without the basis functions, and

Figure 3. (Color online) Primal-Dual VFA



second, we assume full knowledge about all information at each decision point. Then, Step 4d describes the transformation to the new state $S_{t+1}^{n,\text{dual}}$ using the same information that was also used in the primal trajectory ($\hat{\boldsymbol{\omega}}_{t+1}^{n}$). This procedure results in the equivalent outcomes as applying ex post recursion. Finally, Step 5 updates the weights by using decisions and values of the postdecision states of both trajectories that were stored in Step 3c (primal) and Step 4c (dual).

4.4. Feature Selection

Our solution approach relies on features to aggregate the high-dimensional state space. Thereby, features should summarize states of similar values and differentiate states of different values. The design of feature sets needs to be done with care for two reasons. First, too many features impede the learning process as learning many parameters is more difficult than learning just a few relevant ones. Second, the objective in our problem is twofold, and with this, the same set of features may not be equally effective for both objectives as some features are only relevant for either costs or eco-label violations, whereas others may be relevant for both. In the following, we discuss factors that influence the two parts of the objective from which we then derive the features used in our study.

Some problem attributes are equally important to both objectives, for example, a larger distance via truck corresponds to higher costs *and* higher emissions. Thus, the structure of the selected routing paths should be reflected in the features, such as whether an order is routed intermodally using the train, or not. Likewise, it is also important to capture the length of the pre- and postcarriage as they are always conducted by trucks. Features should also consider how long an order blocks the capacity of the train to allow an intermodal routing for future orders that are not yet known when the decision is made for this considered order. Regarding the costs, we need to consider the relative costs for transporting orders via truck, resulting from factors like empty return trips, overtime, or own/rented trucks. We refer to orders whose direct truck transportation is above average as costly orders. Considering all these factors, we design three feature sets for our study: one focusing on eco-labels (*VFA EL*), one focusing on cost (*VFA CO*), and a combination of both (*VFA All*). Table 2 shows a list of the features for each set. The first column shows the feature number, the second states the abbreviated names of the features, the third verbally describes the features, the fourth states how to calculate the value of the features to integrate it in our MIP, and the last three columns indicate whether a feature is included in a feature set (marked by x), or not (marked by —).

The first six features in the table explicitly relate to the orders' requested eco-labels λ_o. For this, we define A, B, and C orders as orders requesting eco-labels A, B, or C, respectively. With this, Features 1–3 count the number of A, B, and C orders that are routed directly using only the truck (road only), and Features 4–6 count the number of A, B, and C orders that are routed intermodally using the train. For the Features 1–6, we consider a feature for each possible combination of an order's label λ_o and load q_o. However, this is not shown in the table for reasons of readability. Features 7–9 relate to the cost structure by counting the number of costly orders that are routed either directly or intermodally. Here, Feature 9 puts a special emphasis on whether a costly order is routed until one of the last three terminals or not, as this comes along with using the train for a relatively long main-carriage compared with orders that use the train only for a few stops. Features 10–13 are relevant for both objectives. Feature 10

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Table 2. Feature Sets Used in Our Study

Number	Name	Description	Calculation	VFA EL	VFA CO	VFA All
1	A-FTLs RO[a]	Number of FTLs, with requested eco-label A, routed road-only	$\sum_{o\in\mathcal{O}_t:\lambda_o=A}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p=\emptyset} r_t(p)$	x	—	x
2	B-FTLs RO[a]	Number of FTLs, with requested eco-label B, routed road-only	$\sum_{o\in\mathcal{O}_t:\lambda_o=B}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p=\emptyset} r_t(p)$	x	—	x
3	C-FTLs RO[a]	Number of FTLs, with requested eco-label C, routed road-only	$\sum_{o\in\mathcal{O}_t:\lambda_o=C}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p=\emptyset} r_t(p)$	x	—	x
4	A-FTLs IM[a]	Number of FTLs, with requested eco-label A, routed intermodally	$\sum_{o\in\mathcal{O}_t:\lambda_o=A}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p\neq\emptyset} r_t(p)$	x	—	x
5	B-FTLs IM[a]	Number of FTLs, with requested eco-label B, routed intermodally	$\sum_{o\in\mathcal{O}_t:\lambda_o=B}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p\neq\emptyset} r_t(p)$	x	—	x
6	C-FTLs IM[a]	Number of FTLs, with requested eco-label C, routed intermodally	$\sum_{o\in\mathcal{O}_t:\lambda_o=C}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p\neq\emptyset} r_t(p)$	—[b]	—	—[b]
7	Costly FTLs RO	Number of costly-FTLs routed road-only	$\sum_{o\in\mathcal{O}_t:a_o=1}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p=\emptyset} r_t(p)\cdot q_o$	—	x	x
8	Costly FTLs IM	Number of costly-FTLs routed intermodally	$\sum_{o\in\mathcal{O}_t:a_o=1}\sum_{p\in\mathcal{P}_t^o:\mathcal{T}_p\neq\emptyset} r_t(p)\cdot q_o$	—	x	x
9	Costly FTLs IM3	Number of costly-FTLs routed intermodally until one of the last three terminals	$\sum_{o\in\mathcal{O}_t:a_o=1}\sum_{p\in\mathcal{P}_t^o:t_p^2\geq\lvert\mathcal{T}\rvert-3} r_t(p)\cdot q_o$	—	x	x
10	FTLs RO	Number of FTLs routed road-only	$\sum_{p\in\mathcal{P}_t:\mathcal{T}_p=\emptyset} r_t(p)\cdot q_o$	—[b]	x	—[b]
11	Terminal	Number of FTLs that are routed intermodally until each terminal	$\sum_{p\in\mathcal{P}_t:t_p^2=t} r_t(p)\cdot q_o,\ \ \forall t\in\mathcal{T}$	x	x	x
12	Cost-ratio	Sum of an order's chosen path cost-ratio for orders routed intermodally times its load size	$\sum_{p\in\mathcal{P}_t:\mathcal{T}_p\neq\emptyset} c_p^r\cdot q_o$	x	x	x
13	Constant	Constant to model the intercept	1	x	x	x

[a]Additionally differentiated per possible order load q_o.
[b]Not considered as this information is already covered by other features.

counts the number of orders that are routed road only, and Feature 11 counts the number of orders that are routed intermodally until each terminal. Feature 12 is the cost-ratio of an order's chosen path and calculated by dividing the path's total costs by the order's road-only costs for a direct routing. This ratio also serves as an indicator for a path's relative length of the pre- and postcarriage. Finally, Feature 13 is a constant to model the intercept. Feature 6 (C-orders IM) is not used in any of the feature sets, and Feature 10 is not included in *VFA EL* and *VFA All* as their information is already covered through other features. For example, Feature 10 is also expressed through the sum of Features 1–3.

All the features are further differentiated between mustGo and mayGo orders, which is a common way of classifying orders in the transportation literature (Mes and Rivera 2017). Thus, every feature (except for the constant) is included twice: once as mustGo and once as mayGo feature. A mustGo order is defined as an order for which an intermodal routing is not possible at future decision points; that is, if we do not decide to route this order intermodally via terminal t (i.e., at the currently considered decision point), it must be routed using the truck only (direct routing). In contrast, a mayGo order can also be routed intermodally at at least one of the future train stops. The differentiation between mustGo and mayGo orders is helpful for the algorithm to identify critical orders with respect to their routing options. All features are linear formulations such that they can be directly integrated into the MIP model by using the calculation shown in Table 2.

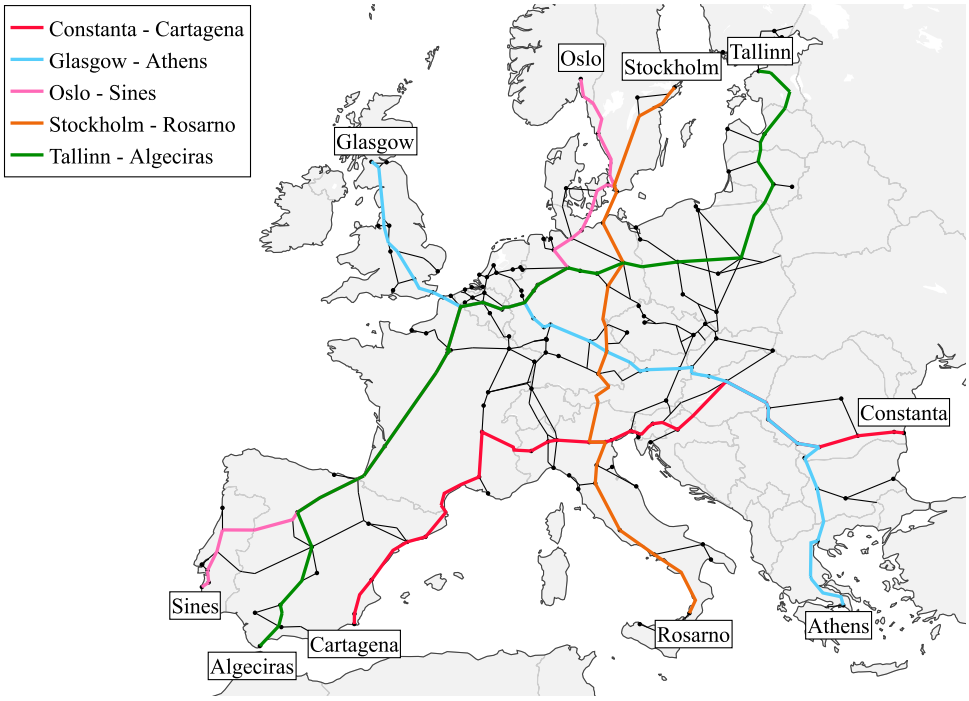
5. Experimental Study

This section presents the experiments that evaluate the proposed solution method and the tradeoff between costs and eco-label violations in our stochastic and dynamic transportation problem. Section 5.1 describes the test data and the experimental setup. The results regarding different aspects of the objective specific feature sets and of the primal-dual VFA are discussed in Sections 5.2 and 5.3, respectively.

5.1. Test Data and Experimental Design

We consider five rail freight corridors in the European railway network as depicted in Figure 4. The infrastructural

Figure 4. (Color online) Five Rail Service Corridors Considered in Our Experimental Study



data and the trains' routes are based on the Trans-European Transport Network and described in the online appendix. Based on this setting, the number of stops (terminals) and decision points for a particular route is between $T = 15$ (Stockholm to Rosarno) and $T = 28$ (Glasgow to Athens). The considered rail corridors describe constellations in which a train traverses over several days. For example, a train on one of TEN-T's prearranged paths between Norway and Italy traverses about three days (https://info-cip.rne.eu/), and intermodal rail/road orders with an air-line distance of at least 300 kilometers have an average transit time of 1.5 days if they are routed in the TEN-T rail network (Heinold and Meisel 2019). To reflect that trains operate regularly as scheduled services, we assume a default load of 500 ton at each of the stops to take account of contractual demand. The remaining capacity can then be used for short-term orders that are modeled as follows. An order's origin and destination location is a randomly chosen point within a 50-km radius around one of the train's stops. We consider orders that appear on relatively short notice whose origin location is around the train's current or next stop. As intermodal routings are more relevant for orders with relatively large direct distances, we assume that an order's destination is around a terminal that is at least four terminals ahead of the train's current stop. To reflect that customers are more interested in using a train for long distances, we assume that the number of orders appearing at the earlier stops of a route is larger compared with later stops. In particular, we consider the case that 30 orders are in expectation at the first stop and that this number is reduced by 3 with each additional stop until it reaches 4. We compute the number of orders using a normal distribution with a standard deviation of four. The size of the orders varies with a discrete uniform distribution between $q_o = 1$ and $q_o = 3$ FTL at 16 tons, and we assume a randomly chosen eco-label A, B, or C, each with a probability of one-third. With these assumptions, it is not possible to route all the considered orders intermodally via the train as it does not have sufficient capacity. Despite this, we do not account for order-specific revenues as we consider a setting in which the same shipper operates both the train and the truck.

To reflect different cost structures among the orders, we select with equal probability if an order is considered a costly order or not and assume twice as high costs for direct transportation for these orders, for example, due to empty travel. Cost rates for truck transportation are based on average values reported between European Union's NUTS2-level regions (Persyn, Díaz-Lanchas, and Barbero 2020). For example, for routing an order from Copenhagen (Denmark) to Hamburg (Germany) we use a rate of 2.46 Euro per FTL-km, and for routing the same order to Rome (Italy), we use a rate of 2.25 Euro per FTL-km. The train operates on a significantly lower cost rate of 0.32 Euro per FTL-km (Forkenbrock 2001, Janic 2007). Emissions are calculated using the EcoTransIT world emission estimation model (EcoTransIT World Initiative 2020). For the trucks, we assume a fix

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emission rate of 70 gCO_2 e per ton-km. For trains, we consider load-dependent emission rates that are linear between four load states at 500, 750, 1,250, and 2,000 tons that result in emission rates of 24.15, 18.78, 13.68, and 10.22 gCO_2 e per ton-km, respectively (Figure 1(b)). Finally, we assume for the train a total capacity of 78 FTL, resulting in a total payload of 1,748 ton for a fully loaded train (including the default load). For simplicity, we use the load as single capacity restriction, whereas the length of the train might also be relevant in practice (European Commissions 2014).

We generally report results for three solution approaches (policies):

1. *Primal-dual VFA*: Our proposed VFA using one of the previously described feature sets (*VFA EL*, *VFA CO*, *VFA All*). For reasons of readability, we refer to this approach simply as VFA. For tuning, we conduct $n = 200$ learning iterations using $\beta = 0.5$ as the default primal-dual learning parameter. To analyze the stability of our learning, we repeat this procedure five times each with a different random seed. Notably, the five VFAs perform very similar, that is, the learning process is quite stable.
2. *Single stage*: A simple heuristic solving the problem just like VFA but ignoring any future impact; that is, it solves the MIP decision problem in a point t by setting the weight of all features to zero.
3. *Dual*: A solution obtained from applying the same logic as for calculating the dual trajectory in the VFA algorithm. With this, it constitutes the lower bound for a considered realization.

We also considered rule-based decision policies, such as routing all orders with eco-label A or B via the train and all orders with eco-label C directly via truck, but preliminary tests showed an inferior performance even when compared with the single-stage heuristic.

We run 100 realizations for each of the five rail corridors and provide aggregated results by reporting an average lower-bound gap (in %) over the 500 realizations. We refer to the online appendix for results per corridor that are very similar under each objective. The lower-bound gap reflects the difference in the objective value between the considered approach (VFA or single stage) and the dual solution and is calculates as follows:

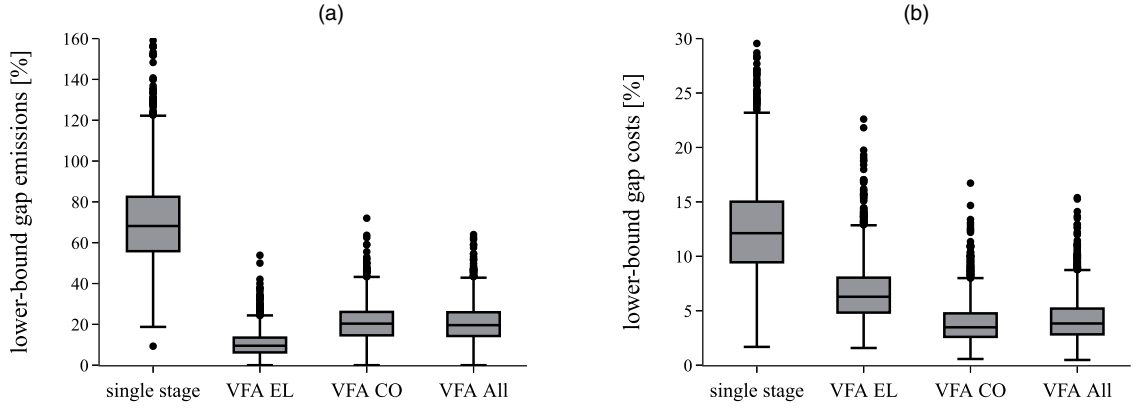
$$\frac{\text{(policy's objective value)} - \text{(dual's objective value)}}{\text{(dual's objective value)}}.$$

Regarding the general fit of our feature sets, we perform an analysis of the coefficient of determination that is described in the online appendix. There, we also show that our basis functions perform well when the test problem realizations are drawn from a slightly different distribution than the learning one. In particular, we provide results for settings in which we fix the size of the orders to one, two, or three FTLs each. We solve all MIPs using IBM CPLEX Optimizer Version 12.9. For our largest rail corridor (Glasgow to Athens), the computing time is below two seconds per decision point for the primal MIP solution (and for the single-stage heuristic) under both objectives and generally below four seconds and half a minute for the dual MIP solution under objectives costs and eco-labels, respectively.

5.2. Value of Objective-Specific Feature Sets

We start by analyzing how different solution approaches perform when striving for a minimization of costs and eco-label violations. Figure 5, (a) and (b), shows the boxplots of the lower-bound gaps when setting cost and eco-labels as objective, respectively, as achieved by the solution approaches single stage, dual, *VFA EL*, *VFA CO*, and *VFA All*. It can be seen that the approaches perform differently for the two measures. The median gap of the eco-label's violation level is 9.5% under eco-label feature set *VFA EL*, which is much lower than the median gap of the single-stage heuristic

Figure 5. Results for Objectives Eco-Label Violations and Costs



Notes. (a) Objective eco-label violations ($\alpha = 0$). (b) Objective costs ($\alpha = 1$).

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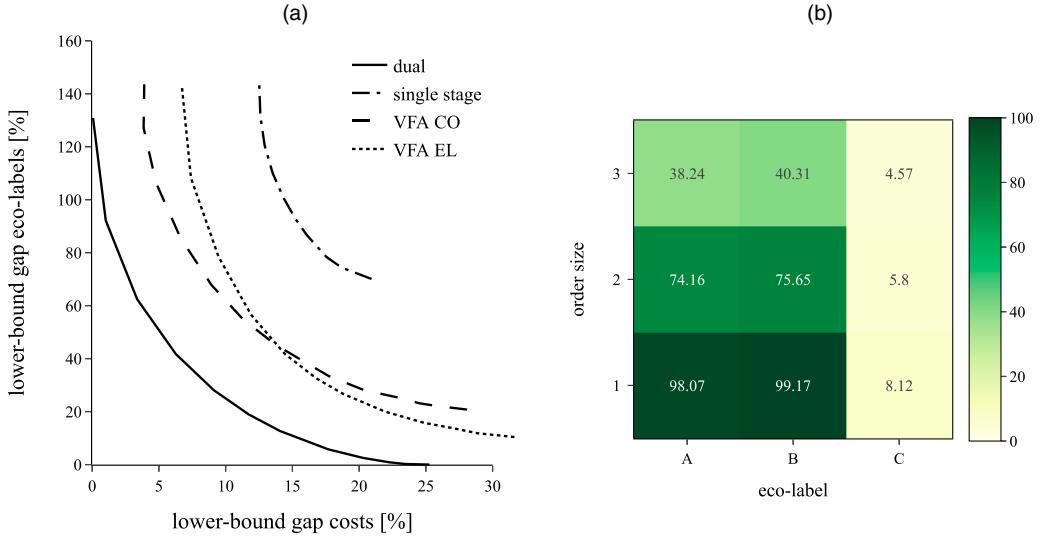
(68.2%) (Figure 5(a)). Similarly, according to the lower-bound gap for costs reported in Figure 5(b), the median is 3.5% using the cost-specific feature set *VFA CO*, and with this, it is considerably lower that the median gap of the single-stage heuristics (12.1%). This shows that the VFA learning process is very helpful in obtaining good solutions for a stochastic dynamic setting, where information to calculate the dual solution is actually not available, and therefore a lower-bound gap of zero constitutes per definition merely a theoretic benchmark.

Comparing the results of the three VFA feature sets, we see that VFA solutions obtained from an objective-specific feature set clearly outperform solutions obtained from VFAs with other feature sets. In particular, the lower quartile (Q1) and the upper quartile (Q3) are remarkably lower for *VFA EL* compared with *VFA CO* under objective eco-labels (Q1: 6.3% versus 14.6%; Q3: 13.5% versus 26.1%). Likewise, they are remarkably lower for *VFA CO* compared with *VFA EL* under objective costs (Q1: 2.6% versus 4.8%; Q3: 4.8% versus 8.0%). Regarding *VFA All*, we see similar results as for *VFA CO*. This is because aggregating features that are specific to eco-labels reflects aspects that are generally important in our decision making (e.g., the number of intermodally routed orders is equal to the sum of intermodally routed A, B, and C orders), whereas this is not true for the features specific to costs. Thus, using all features is not necessarily the best strategy in our multiobjective decision problem as the nonsignificant features impede the learning. We therefore present subsequent results using the objective specific features sets.

To underpin the findings, Figure 6(a) depicts the pareto frontiers of our approaches demonstrating the general tradeoff between the two objectives. Here, we set parameter α to values between zero and one representing different weightings for objectives eco-labels and costs. The lower-bound gap is calculated as follows: we solve the problem with $\alpha = \{0, 1\}$ and then evaluate the costs and eco-label violations of an approach using the values from these pure objectives. We scaled the axes differently for the purpose of presentation. In the figure, results with the lowest eco-label violations ($\alpha = 0$) correspond to low values on the vertical axis and results with the lowest costs ($\alpha = 1$) correspond to low values on the horizontal axis. It can be seen that the VFA solutions lie between the dual solutions obtained ex post and the solutions under the single-stage heuristic in all of the considered α-settings. Generally, less eco-label violations come along with an increase in the cost and vice versa. Thus, the results clearly illustrates the tradeoff between the two objectives that is particular relevant for α values close to zero or one. For example, the VFA curves are very steep at low values for the cost gap, indicating that a relatively small improvement in the cost gap corresponds to a large increase in the eco-label gap. Even the dual solutions show this strong tradeoff. For example, pursuing a pure eco-label objective results in costs that are about 25% larger (horizontal intercept) compared with the pure cost objective.

To better understand the corresponding structural changes of solutions, we now analyze the intermodal share, that is, the percentage of orders that are routed intermodally. This share is on average 47% under the dual solution approach, 49% under *VFA CO*, 51% under *VFA EL*, and 57% under the single-stage heuristic. This means that the overall intermodal share is more or less similar in all three approaches. However, this is different when looking at the intermodal share of

Figure 6. (Color online) Pareto Frontier and Intermodal Share for VFA EL



Notes. (a) Pareto frontier for varying weights of the two objectives. (b) Intermodal share per order size and eco-label for *VFA EL* under objective eco-labels.

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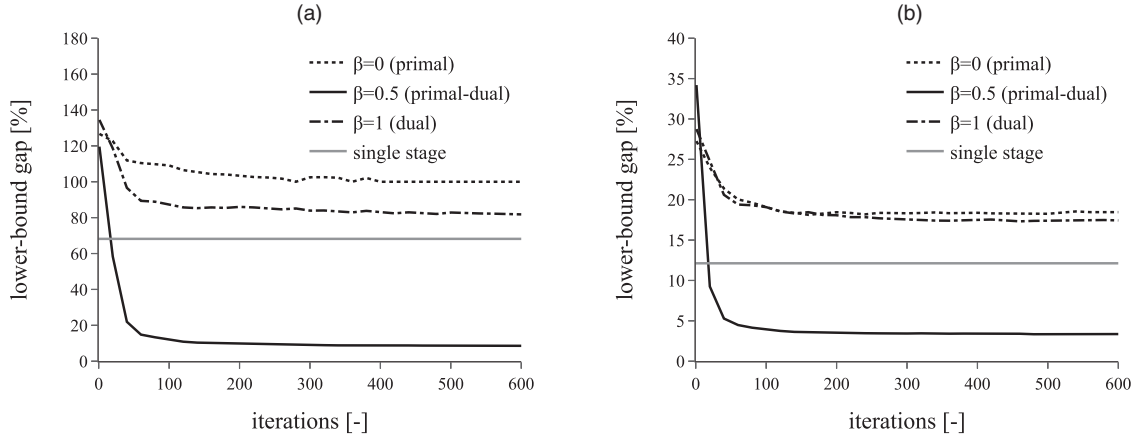
the orders per requested eco-label (A, B, and C orders) and per order size (one, two, and three orders). In this context, A, B, and C orders refer to those orders that respectively request the eco-labels A, B, and C for their shipping (λ_o) and one, two, and three orders refer to those orders whose respective size q_o is equal to one, two, and three FTL. Figure 6(b) shows the intermodal share per eco-label (horizontal axis) and per order size (vertical axis). The share for *VFA EL* is shown under objective eco-labels and the results are depicted as a heatmap in which a darker color corresponds to a higher intermodal share. We refer to the online appendix for the heatmap of the dual solution, which is noticeable similar compared with *VFA EL*, and for the heatmaps of the single-stage heuristic and *VFA CO*, which are noticeable different compared with *VFA EL*.

The results reveal that there are substantial differences between the intermodal share for the possible combinations of the orders' eco-labels and sizes. Two findings stick out. First, the intermodal share is remarkably higher for smaller orders, and second, the intermodal share is remarkably higher for A and B orders than it is for C orders. The first finding is reasonable as it is easier to integrate smaller orders in the train's capacity planning compared with larger orders. The second finding is reasonable as these orders are more sensitive toward their total emissions, and it is therefore beneficial to route them intermodally. Consequently, routing some of the C orders intermodally is then done only to fill up the train (and thus reduce its emission rate for accompanying orders) or due to lower overall costs. Either way, our primal-dual VFA successfully identifies which of these orders should be routed intermodally and which of them should not. This once again demonstrates the benefits from using VFA with features related to eco-labels as this allows the approach to learn the impact of the labels on the appropriate routing decisions. Overall, we conclude that it is important to design feature sets such that they are aligned to the influencing factors of the considered objective.

5.3. Value of Primal-Dual VFA

This section presents results regarding our primal-dual VFA. We start by showing the learning curves for objectives eco-labels (using feature set *VFA EL*) and costs (using feature set *VFA CO*) in Figure 7, (a) and (b), respectively. We show learning curves for three settings of our primal-dual learning parameter β: (i) $\beta = 0$ means that we only use the primal trajectory for updating our VFA, (ii) $\beta = 1$ means that we only use the dual trajectory for updating our VFA, and (iii) $\beta = 0.5$ means that both trajectories are used to the same extent for updating our VFA. The horizontal gray line corresponds to the single-stage heuristic that does not involve learning at all. For the purpose of presentation, we show the lower-bound gap after every 20th iteration and initialize the learning curves at $n = 1$. The horizontal axis depicts the number of iterations where more iterations refer to a longer learning process. The learning curves show the classical behavior of a fast initial learning and a convergence at around $n = 200$ iterations for all considered β settings. The primal-dual VFA ($\beta = 0.5$) clearly outperforms the two pure learning strategies ($\beta = 0$ and $\beta = 1$) in terms of performance and speed. That is, using the primal and the dual trajectories together in the estimation of the features weights improves the learning in VFA. In particular, the pure learning strategies are not superior to the single-stage heuristic, and thus, only by applying our novel primal-dual VFA are we able to produce results that outperform a nonanticipating heuristic. The results from Figure 7 also show that objective-specific feature sets *VFA EL* and *VFA CO* show a quite different behavior under the different learning strategies. According to that, the performance of the feature sets

Figure 7. Learning Curves with Different Settings of the Primal-Dual Learning Parameter β for *VFA EL* Under Objective Eco-Labels and for *VFA CO* Under Objective Costs



Notes. (a) Learning curves of *VFA EL* for three β-settings under objective eco-labels. (b) Learning curves of *VFA CO* for three β-settings under objective costs.

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also correlates with how the different trajectories are weighted in the learning process, which points toward a future research direction.

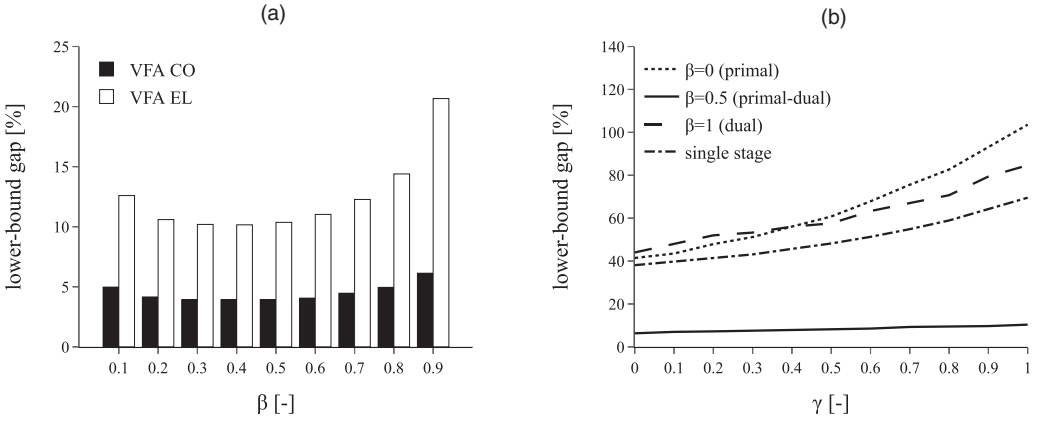
To better understand the influence of the primal-dual learning parameter β, we show in Figure 8(a) the lower-bound gap for both objectives and varying β values between 0.1 and 0.9. The pure learning strategies ($\beta = 0$ and $\beta = 1$) are not reported here as they result in much higher gaps (Figure 7). The gaps in Figure 8(a) correspond to the objective of the respective feature set; for example, gaps for *VFA EL* are reported for objective eco-labels and gaps for *VFA CO* are reported for objective costs. It can be seen that there is a U shape for both objectives. The lowest gaps are observed for β values around 0.5, and they increase almost symmetrically toward lower and higher β values. Thus, the particular chosen weighting of the two trajectories around $\beta = 0.5$ looks quite stable. Regardless of this, it is clear that all the combined learning strategies outperform the pure learning strategies depicted in Figure 7 (even the ones in which one of the trajectories is included through a low weight only). The training of all VFAs requires only a few hours, where training the primal-dual VFAs takes about twice as long as the training of the pure primal or dual VFAs.

One drawback of classical VFA is that its performance generally decreases with an increasing level of stochasticity (Reddy et al. 2020). This is because it gets more difficult for the algorithm to approximate the value function using the primal trajectories, which get more *noisy* in more stochastic settings. We expect our primal-dual VFA to be a mean to make approximations more robust as the dual trajectory is generally less prone to increasing stochasticity. To show this, we compare the performance of the primal-dual VFA with the pure learning strategies and the single-stage heuristic for varying levels of the stochastic information. For this, we use a variation parameter $\gamma \in [0,1]$ that alters the orders' eco-labels. Parameter γ describes the probability that an order's eco-label is determined randomly using the distribution described in Section 5.1, whereas it is A otherwise. Thus, $\gamma = 0$ means that all orders request eco-label A, and $\gamma = 1$ means that each order's eco-label is A, B, or C with an equal probability of one-third (default setting). Figure 8(b) shows the results for the varied γ values (horizontal axis). We report the lower-bound gap under objective eco-labels (vertical axis) using feature set *VFA EL* for the VFA approaches. It can be seen that our primal-dual VFA ($\beta = 0.5$) outperforms both of the pure learning strategies under all the γ settings, confirming its clear dominance in our problem setting. Furthermore, the results show that the single-stage heuristic and the VFA approaches with $\beta = 0$ (only considering the primal trajectory) and with $\beta = 1$ (only considering the dual trajectory) clearly suffer from an increasing variability of the requested eco-labels as is revealed by an increasing lower-bound gap. In contrast, the VFA approach considering both trajectories ($\beta = 0.5$) is quite robust against this variability. This shows that including the primal *and* the dual trajectory into VFA's learning makes the learning less prone for variations in the level of stochasticity.

6. Conclusion and Outlook

In this paper, we investigated the role of order-specific eco-labels in an intermodal stochastic and dynamic transportation problem. The eco-labels serve as an indicator for a customer's environmental cautiousness and constitute an upper bound on each order's allowed emissions. To solve this problem, we proposed a value function approximation with basis functions, where we enhanced the learning by considering two trajectories of the same iteration. In particular, we used a primal trajectory resulting from the "classical" forward decisions

Figure 8. Results for Varying the Primal-Dual Parameter β and the Eco-Label Stochasticity Parameter γ



Notes. (a) Gap for *VFA CO* (for cost objective, $\alpha = 0$) and *VFA EL* (for eco-label objective, $\alpha = 1$), for varying values of the primal-dual learning parameter β. (b) Results for *VFA EL* under objective eco-labels for varying values of eco-label stochasticity parameter γ.

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and a dual trajectory resulting recursively from an ex post optimization problem. There are several opportunities for future work. It might be interesting to extend the problem toward dimensions like multiple trains, additional modes for the main carriage (e.g., barges), multiple periods, or uncertainty regarding the schedules. Furthermore, as the VFA can evaluate the potential cost of a customer ordering a specific eco-label, our problem and method may be extended toward guiding customer selection, for example, via dynamic pricing or slotting. We have addressed a multiobjective problem considering the minimization of costs and the number of orders violating their requested eco-label. For this, we have designed feature sets that focus on those factors that influence eco-labels, costs, or both. Our experiments in a European rail corridor showed that objective-specific feature sets outperform features sets that do not relate to the weight of the two parts of the objective. Here we see the research opportunity of dynamically selecting features according to a (weighted) objective.

We also demonstrated in the experiments that our primal-dual VFA improves the learning both in terms of quality and speed compared with VFAs that solely use either the primal trajectory or the dual trajectory. We showed that this is because the combination of the two works as kind of exploration strategy as it evaluates yet another trajectory of the observed information. In the experiments, we set an a priori relation to specify how to weight the two trajectories. For future work, it might be interesting to analyze the impact of a dynamic weighting of the two trajectories in the learning process (e.g., higher weight at the beginning, lower weight at the end). Finally, as our primal-dual VFA is a general methodological contribution to approximate dynamic programming, it would be worth looking into how to apply it to problems not related to transportation.

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References

Agatz N, Fan Y, Stam D (2021) The impact of green labels on time slot choice and operational sustainability. *Production Oper. Management* 30(7):2285–2303.

Agatz N, Hewitt M, Thomas BW (2021) "Make no little plans": Impactful research to solve the next generation of transportation problems. *Networks* 77(2):269–286.

Archetti C, Peirano L, Speranza MG (2022) Optimization in multimodal freight transportation problems: A survey. *Eur. J. Oper. Res.* 299(1):1–20.

Bauer J, Bektaş T, Crainic TG (2010) Minimizing greenhouse gas emissions in intermodal freight transport: An application to rail service design. *J. Oper. Res. Soc.* 61(3):530–542.

Baykasoğlu A, Subulan K (2016) A multi-objective sustainable load planning model for intermodal transportation networks with a real-life application. *Transportation Res., Part E Logist. Transportation Rev.* 95:207–247.

Bellman R (1957) *Dynamic Programming* (Princeton University Press, Princeton, NJ).

Brown DB, Smith JE, Sun P (2010) Information relaxations and duality in stochastic dynamic programs. *Oper. Res.* 58(4-part-1):785–801.

Chen K, Yang Z, Notteboom T (2014) The design of coastal shipping services subject to carbon emission reduction targets and state subsidy levels. *Transportation Res., Part E Logist. Transportation Rev.* 61:192–211.

Crainic TG, Kim KH (2007) Intermodal transportation. *Handbook Oper. Res. Management Sci.* 14:467–537.

Delbart T, Molenbruch Y, Braekers K, Caris A (2021) Uncertainty in intermodal and synchromodal transport: Review and future research directions. *Sustainability* 13(7):3980.

Demir E, Bektaş T, Laporte G (2011) A comparative analysis of several vehicle emission models for road freight transportation. *Transportation Res. Part D Transportation Environment* 16(5):347–357.

Demir E, Burgholzer W, Hrušovský M, Arikan E, Jammernegg W, Van Woensel T (2016) A green intermodal service network design problem with travel time uncertainty. *Transportation Res. Part B: Methodological* 93:789–807.

Dumouchelle J, Frejinger E, Lodi A (2021) Can machine learning help in solving cargo capacity management booking control problems? Preprint, submitted January 29, https://arxiv.org/abs/2102.00092.

EcoTransIT World Initiative (2020) Ecological transport information tool for worldwide transports (update 2020). Accessed January 7, 2022, https://www.ecotransit.org/wordpress/wp-content/uploads/20210531_Methodology_Report_EcoTransIT_World.pdf.

European Commissions (2014) Core network corridors progress report of the European coordinators. Accessed January 7, 2022, https://ec.europa.eu/transport/infrastructure/tentec/tentec-portal/site/brochures_images/CorridorsProgrReport_version1_2014.pdf.

European Environment Agency (2021) Greenhouse gas emissions from transport in Europe. Accessed January 7, 2022, https://www.eea.europa.eu/ims/greenhouse-gas-emissions-from-transport.

Forkenbrock DJ (2001) Comparison of external costs of rail and truck freight transportation. *Transportation Res. Part A Policy Practice* 35(4):321–337.

George AP, Powell WB (2006) Adaptive stepsizes for recursive estimation with applications in approximate dynamic programming. *Machine Learn.* 65(1):167–198.

George AP, Powell WB, Kualkarni SR (2008) Value function approximation using multiple aggregation for multiattribute resource management. *J. Machine Learn. Res.* 9:2079–2111.

Guo W, Atasoy B, Negenborn R (2022) Global synchromodal shipment matching problem with dynamic and stochastic travel times: A reinforcement learning approach. *Ann. Oper. Res.*, https://doi.org/10.1007/s10479-021-04489-z.

Guo W, Atasoy B, van Blokland WB, Negenborn RR (2021a) Anticipatory approach for dynamic and stochastic shipment matching in hinterland synchromodal transportation. *Flexible Service Manufacturing J.* 34:483–517.

Guo W, Atasoy B, van Blokland WB, Negenborn RR (2021b) Global synchromodal transport with dynamic and stochastic shipment matching. *Transportation Res., Part E Logist. Transportation Rev.* 152:102404.

Heinold A (2020) Comparing emission estimation models for rail freight transportation. *Transportation Res. Part D Transportation Environment* 86:102468.

Heinold A, Meisel F (2018) Emission rates of intermodal rail/road and road-only transportation in Europe: A comprehensive simulation study. *Transportation Res. Part D Transportation Environment* 65:421–437.

Heinold A, Meisel F (2019) Emission oriented vs. time oriented routing in the European intermodal rail/road freight transportation

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network. Bierwirth C, Kirschstein T, Sackmann D, eds. *Logistics Management* (Springer International Publishing, Cham, Switzerland), 188–202.

Heinold A, Meisel F (2020) Emission limits and emission allocation schemes in intermodal freight transportation. *Transportation Res., Part E Logist. Transportation Rev.* 141:101963.

Hrušovský M, Demir E, Jammernegg W, Van Woensel T (2020) Real-time disruption management approach for intermodal freight transportation. *J. Clean Production* 280:124826.

Janic M (2007) Modelling the full costs of an intermodal and road freight transport network. *Transportation Res. Part D Transportation Environment* 12(1):33–44.

Jiang DR, Powell WB (2015) An approximate dynamic programming algorithm for monotone value functions. *Oper. Res.* 63(6):1489–1511.

Jiang DR, Al-Kanj L, Powell WB (2020) Optimistic Monte Carlo tree search with sampled information relaxation dual bounds. *Oper. Res.* 68(6):1678–1697.

Jiang J, Zhang D, Meng Q, Liu Y (2020) Regional multimodal logistics network design considering demand uncertainty and CO2 emission reduction target: A system-optimization approach. *J. Clean Production* 248:119304.

Kirschstein T, Heinold A, Behnke M, Meisel F, Bierwirth C (2022) Eco-labeling of freight transport services: Design, evaluation, and research directions. *J. Industrial Ecology* 26(3):801–814.

Lam JSL, Gu Y (2016) A market-oriented approach for intermodal network optimisation meeting cost, time and environmental requirements. *Internat. J. Production Econom.* 171:266–274.

Levin Y, Nediak M, Topaloglu H (2012) Cargo capacity management with allotments and spot market demand. *Oper. Res.* 60(2):351–365.

Levina T, Levin Y, McGill J, Nediak M (2011) Network cargo capacity management. *Oper. Res.* 59(4):1008–1023.

Li L, Negenborn RR, De Schutter B (2015) Intermodal freight transport planning: A receding horizon control approach. *Transportation Res., Part C Emerging Tech.* 60:77–95.

Lorenzo-Toja Y, Vázquez-Rowe I, Amores MJ, Termes-Rifé M, Marín-Navarro D, Moreira MT, Feijoo G (2016) Benchmarking wastewater treatment plants under an eco-efficiency perspective. *Sci. Total Environment* 566:468–479.

Maxwell MS, Restrepo M, Henderson SG, Topaloglu H (2010) Approximate dynamic programming for ambulance redeployment. *INFORMS J. Comput.* 22(2):266–281.

Mes MRK, Iacob ME (2016) Synchromodal transport planning at a logistics service provider. Zijm H, Klumpp M, Clausen U, Hompel Mt, eds. *Logistics and Supply Chain Innovation: Bridging the Gap Between Theory and Practice* (Springer International Publishing, Cham, Switzerland), 23–36.

Mes MRK, Rivera AP (2017) Approximate dynamic programming by practical examples. Boucherie RJ, van Dijk NM, eds. *Markov Decision Processes in Practice* (Springer International Publishing, Cham, Switzerland), 63–101.

Miller J, Nie YM, Liu X (2020) Hyperpath truck routing in an online freight exchange platform. *Transportation Sci.* 54(6):1676–1696.

Persyn D, Díaz-Lanchas J, Barbero J (2020) Estimating road transport costs between and within European Union regions. *Transportation Policy* 124:33–42.

Piecyk M, Cullinane S, Edwards J (2012) Assessing the external impacts of freight transport. McKinnon A, Browne M, Whiteing A, eds. *Green Logistics: Improving the Environmental Sustainability of Logistics*, 2nd ed. (Kogan Page Limited, London), 31–50.

Powell WB (2009) What you should know about approximate dynamic programming. *Naval Res. Logist.* 56(3):239–249.

Powell WB (2011) *Approximate Dynamic Programming: Solving the Curses of Dimensionality*, 2nd ed. (John Wiley & Sons, Hoboken, NJ).

Powell WB (2021) *Reinforcement Learning and Stochastic Optimization* (John Wiley & Sons, Hoboken, NJ).

Railnet Europe (2021) Rail sector agrees on a common vision on digital capacity management. January 7, 2022, https://rne.eu/news/rail-sector-agrees-on-a-common-vision-on-digital-capacity-management/.

Reddy GT, Reddy MPK, Lakshmanna K, Kaluri R, Rajput DS, Srivastava G, Baker T (2020) Analysis of dimensionality reduction techniques on big data. *IEEE Access* 8:54776–54788.

Resat HG, Turkay M (2019) A discrete-continuous optimization approach for the design and operation of synchromodal transportation networks. *Comput. Industrial Engrg.* 130:512–525.

Rivera AEP, Mes MR (2017a) Anticipatory freight selection in intermodal long-haul round-trips. *Transportation Res., Part E Logist. Transportation Rev.* 105:176–194.

Rivera AEP, Mes MRK (2017b) Scheduling drayage operations in synchromodal transport. Bektaş T, Coniglio S, Martinez-Sykora A, Voß S, eds. *Computational Logistics* (Springer International Publishing, Cham, Switzerland), 404–419.

Rivera AEP, Mes MR (2019) Integrated scheduling of drayage and long-haul operations in synchromodal transport. *Flexible Service Manufacturing J.* 31(3):763–806.

Rudi A, Fröhling M, Zimmer K, Schultmann F (2016) Freight transportation planning considering carbon emissions and in-transit holding costs: A capacitated multi-commodity network flow model. *EURO J. Transportation Logist.* 5(2):123–160.

Schaul T, Zhang S, LeCun Y (2013) No more pesky learning rates. Dasgupta S, McAllester D, eds. *Proc. 30th Internat. Conf. on Machine Learn.*, vol. 20 (JMLR.org), 343–351.

Shar IE, Jiang D (2020) Lookahead-bounded Q-learning. Daumé H III, Singh A, eds. *Proc. 37th Internat. Conf. on Machine Learn.*, vol. 119 (JMLR.org), 8665–8675.

Smith LN (2017) Cyclical learning rates for training neural networks. *Proc. IEEE Winter Conf. on Applications of Computer Vision* (IEEE, Piscataway, NJ), 464–472.

Steadie Seifi M, Dellaert NP, Nuijten W, Van Woensel T, Raoufi R (2014) Multimodal freight transportation planning: A literature review. *Eur. J. Oper. Res.* 233(1):1–15.

Sun Y, Lang M (2015) Modeling the multicommodity multimodal routing problem with schedule-based services and carbon dioxide emission costs. *Math. Problems Engrg.* 2015:406218.

Topaloglu H, Powell WB (2006) Dynamic-programming approximations for stochastic time-staged integer multicommodity-flow problems. *INFORMS J. Comput.* 18(1):31–42.

Ulmer MW (2020) Dynamic pricing and routing for same-day delivery. *Transportation Sci.* 54(4):1016–1033.

Ulmer M, Savelsbergh M (2020) Workforce scheduling in the era of crowdsourced delivery. *Transportation Sci.* 54(4):1113–1133.

Ulmer MW, Thomas BW (2020) Meso-parametric value function approximation for dynamic customer acceptances in delivery routing. *Eur. J. Oper. Res.* 285(1):183–195.

Ulmer MW, Mattfeld DC, Köster F (2018) Budgeting time for dynamic vehicle routing with stochastic customer requests. *Transportation Sci.* 52(1):20–37.

Ulmer MW, Goodson JC, Mattfeld DC, Hennig M (2019) Offline–online approximate dynamic programming for dynamic vehicle routing with stochastic requests. *Transportation Sci.* 53(1): 185–202.

Ulmer MW, Goodson JC, Mattfeld DC, Thomas BW (2020) On modeling stochastic dynamic vehicle routing problems. *EURO J. Transportation Logist.* 9(2):100008.

Van Heeswijk WJ, Mes MR, Schutten JM (2019) The delivery dispatching problem with time windows for urban consolidation centers. *Transportation Sci.* 53(1):203–221.

Van Heeswijk W, Mes M, Schutten J, Zijm W (2018) Freight consolidation in intermodal networks with reloads. *Flexible Service Manufacturing J.* 30(3):452–485.

Zhao Y, Xue Q, Zhang X (2018) Stochastic empty container repositioning problem with CO2 emission considerations for an intermodal transportation system. *Sustainability* 10(11):4211.